



SAPIENZA
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MIP Modelling

Integer linear programming models
for making optimal decisions

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Contents of the course

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- From the model to Python/Gurobi
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Part II — The problems

- Fifteen numerical models (EX 1–15)
- Assignment and scheduling (7 problems)
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- Production planning (3 problems)
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To model

- Twenty numerical problems (N 1–20)
- Twenty symbolic problems (S 1–20)

The course

- Organisation and exam format
- Notebooks in Colab

Material

Site: fabiofurini.github.io/mip-modelling

Code: github.com/fabiofurini/mip-modelling

The method of the course, in one line

For every problem

model $\rightarrow$ proof of the links $\rightarrow$ instance $\rightarrow$ heuristic (primal bound) $\rightarrow$ dual of the relaxation (dual bound) $\rightarrow$ solution with the solver $\rightarrow$ **additional modelling questions**

- First the model is **built**: variables, objective, constraints. Where two families of variables are linked, one explains what the implication is and how the constraint (or optimality) imposes it.
- Then the optimum is **enclosed** between two numbers: a heuristic gives the bound on one side, a dual solution of the LP relaxation gives the bound on the other.
- Finally the model is **solved**, with Gurobi from Python.

Why the sandwich matters

It is the technique used in practice when the instance is too large to be solved to proven optimality: the two bounds say what the solution in hand is worth.

Notation

- **LP** linear programming; **ILP** integer; **BIP** binary; **MILP** mixed integer.
- Numbered objects with indices over enumerated sets: $j \in \{1, 2, \dots, n\}$ — the notation of the text is already that of the model; rational data ($\mathbb{Q}$), integer counts ($\mathbb{Z}_{\geq 1}$).
- The **bar** marks the values of a *feasible* solution ($\bar{x}_j$), the **tilde** those of an *optimal* one ($\tilde{x}_j$).
- Dual variables: π for capacity constraints, μ for equalities; α, β, γ when more families are needed. Never y for the duals.
- Bounds and values: UB , LB , $z(LP)$ (relaxation without the bounds), $z(LP^+)$ (with the bounds), $z(MILP)$.
- In every model: first the **data** (with their type), then the **variables** (with their count), then objective and constraints.

What a MIP model is

Data, variables, objective, constraints

- The **data** are given: they are not decided. In the model they appear as symbols with their type ($c_j \in \mathbb{Q}_{>0}$, $n \in \mathbb{Z}_{\geq 1}$).
- The **variables** are the decisions. They are declared one family at a time, with the count: "we introduce the following n k binary variables".
- The **objective** is a linear function of the variables.
- The **constraints** are grouped by family, and every family is described in words, with the number of constraints it contains.

The rule that is not negotiable

Every row of the model must be **justified**. A constraint nobody can explain is a constraint nobody can fix.

Four classes, one language

Class	Variables	Example in the course
LP	all continuous	(the Operations Research Lab)
ILP	all integer	EX 5, funds in lots
BIP	all binary	EX 1, the van
MILP	mixed	EX 6, vehicles with a minimum quantity

The point

The class changes, the language does not: data, variables, objective, constraints. But what can be *said* about the solution changes radically — and so does the cost of finding it.

Why integrality matters

$$\begin{array}{ll}\max & x_1 + x_2 \\ \text{s.t.} & 2x_1 + 2x_2 \leq 3 \\ & x_1, x_2 \in \{0, 1\}\end{array}$$

- The LP relaxation replaces $x \in \{0, 1\}$ with $0 \leq x \leq 1$ and is worth $z(LP^+) = 3/2$.
- That value is attained by **infinitely many** optimal solutions — the whole segment $x_1 + x_2 = 3/2$ inside the square.
- Rounding is not an approximation: it is **a mistake**. The integer optimum is 1, not 2, and no rounding finds it by chance.

To remember

The value of the relaxation is not "almost" the optimum of the MILP: it is a *bound*. The difference $z(MILP) - z(LP)$ has a name — the price of integrality — and in some problems of the course it is 30%.

The sandwich of the bounds

The chain that holds the whole course together

$$LB \leq z(LP) \leq z(MILP) \leq UB \quad (\text{minimisation problem})$$

- UB comes from a **feasible solution**: any feasible solution is a bound from its own side.
- LB comes from a **feasible dual solution** of the relaxation: by weak duality, *always*, with no assumptions.
- In a **maximisation** problem the two roles swap: the heuristic gives a lower bound, the dual an upper bound.

What the solver does with them

Branch-and-bound does exactly this, at every node: a bound from the relaxation and one from a solution found. When the two meet, it is done.

Relaxations, duality and bounds

What a relaxation is

- A relaxation enlarges the feasible region: every solution of the MILP stays feasible, so its optimum is a bound.
- The regions include each other the right way round:

$$X_{MILP} \subseteq X_{LP^+} \subseteq X_{LP}.$$

- $z(LP)$ is the relaxation **without the bounds** ($x_j \geq 0$ only): it is the one whose dual the course writes by hand.
- $z(LP^+)$ is the relaxation **with the bounds** ($0 \leq x_j \leq 1$): it is what the solver does with `relax()`, and it is tighter.

There is no "dual of the MILP"

The dual one writes is always that of the *relaxation* named. A MILP has no linear dual, and the gap $z(MILP) - z(LP)$ is precisely what is missing.

The primal/dual conversion table

In the primal (min)	In the dual (max)
constraint i of direction $\geq$	variable $\pi_i \geq 0$
constraint i of direction $\leq$	variable $\pi_i \leq 0$
equality constraint i	free variable π_i
variable $x_j \geq 0$	constraint j of direction $\leq c_j$
free variable x_j	equality constraint $j, = c_j$

- Dual objective $\max \sum_i b_i \pi_i$; dual constraint j : $\sum_i a_{ij} \pi_i \leq c_j$.
- If the primal is a **maximisation**, all directions flip.

How to read a dual constraint

"The value I attach to the resources consumed by activity j cannot exceed its cost." With this reading every recipe has an economic meaning — and that is what makes it memorable.

Weak and strong duality

Weak duality — this is the one we need

$\sum_i b_i \bar{\pi}_i \leq \sum_j c_j \bar{x}_j$, *always*, with no assumptions. It gives a bound from **any** feasible dual solution, even one built by hand and far from optimal.

Strong duality — this one is a check

If the LP has a finite optimum, $z(D(LP)) = z(LP)$. The optimum of the dual written by hand *must* match $z(LP)$: if it does not, the dual is written wrong. In the course an assert checks it.

$$\sum_{i=1}^m b_i \bar{\pi}_i \leq z(LP) \leq z(MILP)$$

A dual bound of the LP is a bound for the MILP too, from the same side.

Three recipes for building a dual solution by hand

1. **Zero everything but one family.** The remaining ones are pushed to the largest value compatible with the dual constraints. In 7.1: $\bar{\pi} = 0$ and $\bar{\mu}_j = \min_m c_{jm}$ — "every job costs at least its cheapest machine".
2. **Constructive heuristic on the constraints.** The primal constraints are scanned one by one, the corresponding dual variable is raised until the first opposing dual constraint is tight, and the residuals are updated.
3. **The best ratio.** With a single capacity constraint in a maximisation, $\bar{v} = \max_j p_j/w_j$ is feasible and gives the bound $b \bar{v}$: "all the capacity used at the best efficiency".

The only thing that makes the bound valid

The solution built must be **checked feasible**, on every dual constraint, one by one. An unchecked dual value is not a bound.

A certificate that does not hold: EX 5

$$\begin{array}{ll}\max & 2x_1 + 5x_2 + 2x_3 + 4x_4 \\ \text{subject to} & 12x_1 + 15x_2 + 18x_3 + 25x_4 \leq 100 \\ & -x_1 + x_2 - x_3 - x_4 \leq 0 \\ & x_1, x_2, x_3, x_4 \in \mathbb{Z}_{\geq 0}\end{array}$$

Dual: $\min 100\alpha$ with one row per fund, $c_j\alpha \pm \beta \geq p_j$.

It comes naturally to propose $\bar{\alpha} = 5/32$, $\bar{\beta} = 1/8$:

$$12 \cdot \frac{5}{32} - \frac{1}{8} = \frac{56}{32} = \frac{7}{4} < 2.$$

Violated by 1/4: not a feasible solution, and $100 \cdot 5/32$ is not a bound.

The correct recipe

$\bar{\beta} = 0$ and $\bar{\alpha} = \max_j p_j/c_j = 1/6$ — "a million is worth what it returns in the better fund" — giving $UB = 50/3 \approx 16.67$, against $z(MILP) = 16$: the difference $2/3$ is the price of integrality.

LP duals are not the marginal prices of the MILP

The most expensive mistake of the chapter

In an LP the dual π_i is the marginal price: one more unit of resource i is worth π_i . In a MILP it is not. The integer optimum is a step function of the availability: between steps one more unit is worth nothing, on a step it is worth a lot.

- The π_i values the solver returns are those of the *relaxation*, not of the MILP.
- To know what one more unit of a resource is really worth in a MILP there is only one way: **solve it again** with the data changed.

From the model to Python/Gurobi

The model, one family per block

- One solver only: **Gurobi from Python**. One `addConstrs` per family, named after the label of the model.
- The name is not decoration: it is what lets one read the duals and match code and notes line by line.

The protocol of the course, from start to end

1. the model is written and solved;
2. Status is read — **always before** `ObjVal`, which on an unsolved model raises an error;
3. the heuristic is built and its feasibility *checked*;
4. the dual is written, a solution is built by hand and checked;
5. LB , $z(LP)$, $z(LP^+)$, $z(MILP)$, UB are compared in a table.

Reproducibility

```
python3 python/run_all.py regenerates data, figures and notebooks;  
python3 python/check_numbers.py checks every number quoted in the notes.
```

Tolerances

- A solver works in floating point: " $x_j = 1$ " is read `x[j].X > 0.5`, never `== 1`.
- A "satisfied" constraint is satisfied within a tolerance. In bound comparisons the course uses 10^{-6} .
- Fractions are reported as fractions (25/4, 50/3), not as truncated decimals: that is how exactness is not lost between code and text.

The "no invented number" rule

Every number appearing in the notes, in the slides and on the site comes from a script, and is checked by an assert. If a number changes, the check fails and the CI does not publish.

Constructive heuristics

What a constructive heuristic is

- It builds a solution **one piece at a time**, never backtracking, following a declared rule.
- It serves two purposes: giving a bound on the primal side, and giving the student a way of *understanding* the problem before delegating it to the solver.
- It must be run **step by step**, showing the residual availabilities: that is the part graded at the exam.

Rule	Where it is used
next-fit, first-fit, best-fit	packing into containers
the least loaded (LPT)	balancing, EX 12
the best ratio	covering, EX 13
the cheapest completion	lot sizing

When a heuristic fails, and what it gives

- A constructive heuristic can **get stuck**: in EX 9 it places three queens and then finds no free squares. The bound stays valid, but it is weak — 3 against 4.
- It can also **be off by very little**: in EX 6 the gap is 0.2%.
- What it *always* gives is a **feasible** solution: and that must be checked, not assumed.

What it does not give

It does not give the optimum, and it does not say how far it is. Only the bound from the other side says that: which is why the two halves of the course belong together.

Logic and binary variables

From a sentence to a constraint

- A binary variable is a **proposition**: $x_j = 1$ means "project j is funded".
- The logical connectives (AND , OR , NOT) are translated into linear constraints, through conjunctive normal form.
- Every clause becomes *one* linear constraint.

Sentence	Constraint
at most one of $x_1, \dots, x_n$	$\sum_j x_j \leq 1$
at least one	$\sum_j x_j \geq 1$
exactly one	$\sum_j x_j = 1$
if x_1 then x_2	$x_1 - x_2 \leq 0$
x_1 and x_2 not together	$x_1 + x_2 \leq 1$

Check the translation, do not trust it

Mutual exclusion: what it does *not* say

The constraint $x_1 + x_2 \leq 1$ says "not both". It does **not** say "at least one": $x_1 = x_2 = 0$ satisfies it. The converses must be refuted explicitly.

- The safe way to check a translation is to **enumerate**: with few binaries all assignments are listed and the feasible set is checked against the sentence.
- In the course every translation is checked this way, by a script, with an assert.

How much six implications cut

On the ten projects of exercise 2.1: the $2^{10} = 1024$ assignments drop to 234 once all six implications are imposed — less than a quarter. None of them, on its own, cuts more than half.

The fourteen links between variables

The map of the techniques

1. Activation
2. Fixed cost
3. Minimum lot
4. Integer counts
5. Maximum variable
6. Min-max and max-min
7. Absolute value
8. Big-M
9. Precedences
10. If and only if
11. Counting types
12. Alldiff and binary expansion
13. Soft constraints
14. Piecewise linear functions

Every technique in five steps

(a) the link in words; (b) the linear constraints; (c) the proof in both directions, or the exchange argument; (d) how strong the LP relaxation is; (e) the gurobipy line.

Activation and fixed cost

The link

"The quantity q_j can be positive only if activity j is on."

$$q_j \leq c_j y_j, \quad \forall j \in \{1, 2, \dots, m\} \quad (m \text{ constraints}).$$

- **Direction imposed by the constraint:** if $y_j = 0$ then $q_j \leq 0$, that is $q_j = 0$.
- **Direction that follows from optimality:** if $q_j > 0$ then $y_j = 1$ — but only because y_j has a **strictly positive** cost in the objective. With zero cost the correct conclusion is weaker: *there exists* an optimum with $y_j = 0$.

The mistake not to make

Saying "in every optimum" when the coefficient is only ≥ 0 . If the fixed cost can be zero, the statement must be weakened.

Minimum lot: two constraints, one semi-continuous variable

$$\begin{aligned}\ell y_j &\leq q_j, & \forall j \in \{1, 2, \dots, m\} & \quad (m \text{ constraints}), \\ q_j &\leq c_j y_j, & \forall j \in \{1, 2, \dots, m\} & \quad (m \text{ constraints}).\end{aligned}$$

y_j	what the two constraints impose	admissible q_j
0	$q_j \geq 0$ and $q_j \leq 0$	only $q_j = 0$
1	$q_j \geq \ell$ and $q_j \leq c_j$	$\ell \leq q_j \leq c_j$

- Neither is enough on its own: without the second one produces with $y_j = 0$; without the first the minimum is vacuous.
- One needs $\ell \leq c_j$: otherwise $y_j = 1$ is infeasible and the variable is *forced* to zero — a data error the solver does not report.

Big-M: the smallest the data justify

The rule

M is computed **from the data**, and its value is written right after the model: " $M = \sum_j t_j$ is enough".

- A huge M "to be safe" does not make the model more correct: it only **weakens the LP relaxation**.
- With M too large the relaxation leaves y_j near zero while x_j is positive: the bound weakens and branch-and-bound has to work harder.
- In EX 6 the big-Ms are (66, 80, 50, 44, 36): one per type, all limited by the hours of labour. They are not constants of the model, they are **functions of the data of the instance**.

To remember

Data change $\Rightarrow$ big-Ms change. In EX 11, raising the leather from 120 000 to 200 000 grams, the big-Ms go from (141, 200, 171) to (235, 333, 285).

Maximum variable, min-max, absolute value

$$\begin{aligned} z &\geq t_j x_j, \quad \forall j \in \{1, 2, \dots, n\} && (n \text{ constraints}), \\ z &\geq 0. \end{aligned}$$

- The constraints impose $z \geq \max_j t_j x_j$; **optimality** makes it an equality, because z is minimised.
- The absolute value $|u - v|$ in a *minimisation* costs one variable $d \geq 0$ and two constraints, $d \geq u - v$ and $d \geq v - u$.
- In a *maximisation* the same trick fails: a binary and a big-M are needed, because the disjunction has to be imposed, not obtained for free.

"Balanced" is not an objective: it is a word

Minimising the *maximum* of the loads and minimising their *difference* give the same optimal solutions but different values: in EX 12, 9 against 0. Reporting one for the other is a mistake.

If and only if: when the link must hold both ways

- $x_i \leq y$ says "if there is activity then the indicator is on", **not** the converse.
- If a constraint *counts* the indicators — say $\sum_i y_i \geq 2$ — the missing direction makes that constraint **vacuous**: it is enough to switch y on without doing anything.

The case of EX 15, the music school timetable

With the link $x_{dti} \leq y_{di}$ alone the solver puts all the guitar on Monday and all the violin on Tuesday — two days with a single instrument — and declares zero violated preferences against the two real ones. What is also needed is

$$y_{di} \leq \sum_{t \in T} x_{dti}.$$

The rule

When a constraint counts indicators, the link must hold in both directions.

The problems

How Part II is built

- **Fifteen numerical models** (EX 1–15): explicit data, few variables, one technique each. They are read to get one's bearings.
- **Four families** of general problems, with symbolic model, instance, heuristic, dual and additional questions:
 - assignment and scheduling — 7 problems;
 - location and covering — 4 problems;
 - production planning — 3 problems;
 - mixed problems — 9 problems.

The rule for writing models

The **general** model is always symbolic (n, m, c_j); the **instance** model is always written **out in full**, one column per variable. Never a model halfway between the two.

EX 1 — the eight-seat van

$$\begin{array}{llll} \max & 30x_1 & +50x_2 & +80x_3 & +70x_4 \\ \text{s.t.} & 2x_1 & +3x_2 & +4x_3 & +5x_4 \leq 8 \\ & x_1 & +x_2 & +x_3 & +x_4 \leq 2 \\ & & x_2 & & -x_4 \leq 0 \\ & x_1, & x_2, & x_3, & x_4 \in \{0, 1\} \end{array}$$

LB (heuristic)	110
UB (dual by hand)	160
$z(LP)$	160
$z(LP^+)$	140
$z(MILP)$	120

- The dual by hand ($\bar{\beta} = \bar{\gamma} = 0$, $\bar{\alpha} = \max_j p_j/a_j = 20$) is *optimal* for the relaxation without the bounds: "a seat is worth as much as the group that pays best for it".
- Heuristic gap 8.3%; certified gap 41.7%.

Assignment and scheduling — seven problems

	Problem	What it teaches
7.1	minimum-cost assignment	the base family: no link to prove
7.2	machines with a fixed cost	aggregated activation, and its weak relaxation
7.3	job selection	not all jobs have to be done
7.4	parallel jobs	several machines for the same job
7.5	classes with setup	the setup is paid per class, not per job
7.6	classes with a bonus	an "if and only if" with two link constraints
7.7	total tardiness	sequencing with big-M, and the right big-M

Location and covering — four problems

	Problem	What it teaches
8.1	capacitated location	activation with a capacity
8.2	p -median	the number of sites is fixed, not costed
8.3	covering with interference	set covering plus mutual exclusion
8.4	hub with a maximum cost	the maximum variable, and four relaxations

Aggregated against disaggregated (8.4)

The same link can be written aggregated or disaggregated. *Adding* the disaggregated links tightens the relaxation; *replacing* the aggregated one loses the capacity and makes the optimum collapse from 19 to 10. The four relaxations are 1015/78, 79/6, 11, 10.

Production planning and mixed problems

Production — 3 problems

- 9.1 lot sizing with a fixed cost
- 9.2 production and workforce
- 9.3 vehicles with a minimum lot

Here the variables do not say *whether*, they say *how much*: inventory balance, workforce, hiring.

Mixed problems — 9 problems

- 10.1 prizes with two modes
- 10.2 combinatorial auction
- 10.3 diet with a minimum lot
- 10.4 trees and boxes of lights
- 10.5 shipments in boxes
- 10.6 children among summer camps
- 10.7 branches between two companies
- 10.8 tracks among CDs
- 10.9 books among shelves

Where the LP relaxation stops helping

In the mixed problems the relaxation is often worth zero. The bound from below has to be found with **combinatorial** arguments — parity, counting the containers — not with the dual.

The fifteen numerical models

Fifteen numerical models

Explicit data, few variables, one technique per model. Every slide shows the model of the instance written out — one column per variable, as the script generates it — and the five numbers of the sandwich.

- LB and UB are the two sides of the sandwich built *by hand*: the heuristic solution on one side, the dual certificate on the other.
- $z(LP)$ is the relaxation without the bounds, $z(LP^+)$ the one with them, $z(MILP)$ the integer optimum.
- In a minimisation the heuristic gives UB and the dual LB ; in a maximisation the two roles swap.

EX 1 — which groups of tourists to accept

$$\begin{array}{llllll} \max & 30x_1 & +50x_2 & +80x_3 & +70x_4 & \\ \text{subject to} & 2x_1 & +3x_2 & +4x_3 & +5x_4 & \leq 8 \\ & x_1 & +x_2 & +x_3 & +x_4 & \leq 2 \\ & & x_2 & & -x_4 & \leq 0 \\ & x_1, & x_2, & x_3, & x_4 & \in \{0, 1\} \end{array}$$

$$LB = 110 \cdot UB = 160 \cdot z(LP) = 160 \cdot z(LP^+) = 140 \cdot z(MILP) = 120$$

EX 2 — assignment with a capacity in number of lines

$$\begin{array}{ll}
 \min & 10x_{11} + 4x_{12} + 9x_{13} + 7x_{14} + x_{21} + 2x_{22} + 3x_{23} + 10x_{24} + 8x_{31} + 9x_{32} + 10x_{33} + x_{34} \\
 \text{subject to} & x_{11} \qquad \qquad \qquad + x_{21} \qquad \qquad \qquad + x_{31} \qquad \qquad \qquad = 1 \\
 & \qquad x_{12} \qquad \qquad \qquad + x_{22} \qquad \qquad \qquad + x_{32} \qquad \qquad \qquad = 1 \\
 & \qquad \qquad x_{13} \qquad \qquad \qquad + x_{23} \qquad \qquad \qquad + x_{33} \qquad \qquad \qquad = 1 \\
 & \qquad \qquad \qquad x_{14} \qquad \qquad \qquad + x_{24} \qquad \qquad \qquad + x_{34} \qquad \qquad \qquad = 1 \\
 & x_{11} + x_{12} + x_{13} + x_{14} \qquad \qquad \qquad \leq 2 \\
 & \qquad \qquad x_{21} + x_{22} + x_{23} + x_{24} \qquad \qquad \qquad \leq 2 \\
 & \qquad \qquad \qquad \qquad x_{31} + x_{32} + x_{33} + x_{34} \leq 2 \\
 & x_{11}, x_{12}, x_{13}, x_{14}, x_{21}, x_{22}, x_{23}, x_{24}, x_{31}, x_{32}, x_{33}, x_{34} \in \{0, 1\}
 \end{array}$$

$$UB = 13 \cdot LB = 7 \cdot z(LP) = 9 \cdot z(LP^+) = 9 \cdot z(MILP) = 9$$

EX 3 — assignment with more resources than tasks

$$\begin{array}{ll}
 \min & 37x_{11} + 32x_{12} + 33x_{13} + 37x_{14} + 43x_{21} + 33x_{22} + 42x_{23} + 34x_{24} + 33x_{31} + 28x_{32} + 38x_{33} + 30x_{34} \\
 \text{subject to} & x_{11} + x_{12} + x_{13} + x_{14} = 1 \\
 & \phantom{x_{11} + x_{12} + x_{13} + x_{14}} x_{21} + x_{22} + x_{23} + x_{24} = 1 \\
 & \phantom{x_{11} + x_{12} + x_{13} + x_{14} + x_{21} + x_{22} + x_{23} + x_{24}} x_{31} + x_{32} + x_{33} + x_{34} = 1 \\
 & x_{11} \phantom{+ x_{12} + x_{13} + x_{14}} + x_{21} \phantom{+ x_{22} + x_{23} + x_{24}} + x_{31} \leq 1 \\
 & \phantom{x_{11} +} x_{12} \phantom{+ x_{13} + x_{14}} + x_{22} \phantom{+ x_{23} + x_{24}} + x_{32} \leq 1 \\
 & \phantom{x_{11} + x_{12} +} x_{13} \phantom{+ x_{14}} + x_{23} \phantom{+ x_{24}} + x_{33} \leq 1 \\
 & \phantom{x_{11} + x_{12} + x_{13} +} x_{14} \phantom{+ x_{21} + x_{22} + x_{23} + x_{24}} + x_{24} \phantom{+ x_{31} + x_{32} + x_{33}} + x_{34} \leq 1 \\
 & x_{11}, x_{12}, x_{13}, x_{14}, x_{21}, x_{22}, x_{23}, x_{24}, x_{31}, x_{32}, x_{33}, x_{34} \in \{0, 1\}
 \end{array}$$

$$UB = 99 \cdot LB = 93 \cdot z(LP) = 95 \cdot z(LP^+) = 95 \cdot z(MILP) = 95$$

EX 4 — the minimum number of hubs

$$\begin{array}{llllllllll}
 \min & y_1 + y_2 + y_3 + y_4 + y_5 + y_6 + y_7 + y_8 & & & & & & & & \\
 \text{subject to} & y_1 + y_2 & & + y_4 & & & + y_6 & & & \geq 1 \\
 & y_1 + y_2 & & & & & + y_6 & & & \geq 1 \\
 & & & y_3 & & + y_5 & & & & \geq 1 \\
 & y_1 & & & + y_4 & & & & & \geq 1 \\
 & & & y_3 & & + y_5 & & + y_7 & & \geq 1 \\
 & y_1 + y_2 & & & & & + y_6 & & & \geq 1 \\
 & & & & & y_5 & & + y_7 + y_8 & & \geq 1 \\
 & & & & & & & y_7 + y_8 & & \geq 1 \\
 & y_1, & y_2, & y_3, & y_4, & y_5, & y_6, & y_7, & y_8 & \in \{0, 1\}
 \end{array}$$

$$UB = 3 \cdot LB = 3 \cdot z(LP) = 3 \cdot z(LP^+) = 3 \cdot z(MILP) = 3$$

EX 5 — four mutual funds

$$\begin{array}{ll}\max & 2x_1 + 5x_2 + 2x_3 + 4x_4 \\ \text{subject to} & 12x_1 + 15x_2 + 18x_3 + 25x_4 \leq 100 \\ & -x_1 + x_2 - x_3 - x_4 \leq 0 \\ & x_1, x_2, x_3, x_4 \in \mathbb{Z}_{\geq 0}\end{array}$$

$$LB = 16 \cdot \quad UB = \frac{100}{3} \cdot \quad z(LP) = \frac{700}{27} \cdot \quad z(LP^+) = \frac{700}{27} \cdot \quad z(MILP) = 23$$

EX 6 — five types of vehicle

$$\begin{array}{llllll}
 \max & 200x_1 + 250x_2 + 300x_3 + 550x_4 + 700x_5 \\
 \text{subject to} & 2x_1 + 3x_2 + 5x_3 + 6x_4 + 8x_5 & \leq & 1000 \\
 & 30x_1 + 25x_2 + 40x_3 + 45x_4 + 55x_5 & \leq & 2000 \\
 & x_1 & & -10y_1 & \geq & 0 \\
 & & x_2 & & -10y_2 & \geq & 0 \\
 & & & x_3 & & -10y_3 & \geq & 0 \\
 & & & & x_4 & & -5y_4 & \geq & 0 \\
 & & & & & x_5 & & -5y_5 & \geq & 0 \\
 & x_1 & & & -66y_1 & \leq & 0 \\
 & & x_2 & & & -80y_2 & \leq & 0 \\
 & & & x_3 & & & -50y_3 & \leq & 0 \\
 & & & & x_4 & & & -44y_4 & \leq & 0 \\
 & & & & & x_5 & & & -36y_5 & \leq & 0 \\
 & x_1, & x_2, & x_3, & x_4, & x_5 & & & & \in & \mathbb{Z}_{\geq 0} \\
 & & & & & & y_1, & y_2, & y_3, & y_4, & y_5 & \in & \{0, 1\}
 \end{array}$$

$$LB = 25200 \quad \cdot \quad UB = \frac{280000}{11} \quad \cdot \quad z(LP) = \frac{280000}{11} \quad \cdot \quad z(LP^+) = \frac{229000}{9} \quad \cdot \quad z(MILP) = 25250$$

EX 7 — which orders to accept

$$\begin{array}{rcll}
 \max & 2x_1 & +3x_2 & +x_3 -3y_1 -2y_2 \\
 \text{subject to} & 200x_1 +400x_2 +200x_3 & & \leq 1000 \\
 & x_1 & & -3y_1 \leq 0 \\
 & & x_2 & -2y_2 \leq 0 \\
 & & & x_3 -5y_3 \leq 0 \\
 & x_1, & x_2, & x_3 \in \mathbb{Z}_{\geq 0} \\
 & & y_1, & y_2, & y_3 \in \{0, 1\}
 \end{array}$$

$$LB = 4 \cdot UB = 5 \cdot z(LP) = 5 \cdot z(LP^+) = 5 \cdot z(MILP) = 5$$

EX 8 — selection with exact cardinality and non-adjacency

$$\begin{array}{ll}
 \max & 8x_{11} + 6x_{12} + 5x_{13} + 3x_{14} + 7x_{21} + 9x_{22} + 4x_{23} + 6x_{24} + 5x_{31} + 7x_{32} + 8x_{33} + 9x_{34} \\
 \text{subject to} & x_{11} \qquad \qquad \qquad + x_{21} \qquad \qquad \qquad + x_{31} \qquad \qquad \qquad \leq 1 \\
 & \qquad x_{12} \qquad \qquad \qquad + x_{22} \qquad \qquad \qquad + x_{32} \qquad \qquad \qquad \leq 1 \\
 & \qquad \qquad x_{13} \qquad \qquad \qquad + x_{23} \qquad \qquad \qquad + x_{33} \qquad \qquad \qquad \leq 1 \\
 & \qquad \qquad \qquad x_{14} \qquad \qquad \qquad + x_{24} \qquad \qquad \qquad + x_{34} \qquad \qquad \leq 1 \\
 & x_{11} + x_{12} + x_{13} + x_{14} \qquad \qquad \qquad \qquad \qquad \qquad \leq 1 \\
 & \qquad \qquad \qquad x_{21} + x_{22} + x_{23} + x_{24} \qquad \qquad \qquad \leq 1 \\
 & \qquad \qquad \qquad \qquad \qquad \qquad x_{31} + x_{32} + x_{33} + x_{34} \leq 1 \\
 & x_{11} + x_{12} + x_{13} + x_{14} + x_{21} + x_{22} + x_{23} + x_{24} + x_{31} + x_{32} + x_{33} + x_{34} = 2 \\
 & x_{11} + x_{12} \qquad \qquad \qquad + x_{21} + x_{22} \qquad \qquad \qquad + x_{31} + x_{32} \qquad \leq 1 \\
 & \qquad x_{12} + x_{13} \qquad \qquad \qquad + x_{22} + x_{23} \qquad \qquad \qquad + x_{32} + x_{33} \qquad \leq 1 \\
 & \qquad \qquad x_{13} + x_{14} \qquad \qquad \qquad + x_{23} + x_{24} \qquad \qquad \qquad + x_{33} + x_{34} \leq 1 \\
 & x_{11}, \quad x_{12}, \quad x_{13}, \quad x_{14}, \quad x_{21}, \quad x_{22}, \quad x_{23}, \quad x_{24}, \quad x_{31}, \quad x_{32}, \quad x_{33}, \quad x_{34} \in \{0, 1\}
 \end{array}$$

$$LB = 18 \cdot UB = 18 \cdot z(LP) = 18 \cdot z(LP^+) = 18 \cdot z(MILP) = 18$$

EX 9 — the largest number of queens

$$\begin{array}{ll}
 \max & x_{11}+x_{12}+x_{13}+x_{14}+x_{21}+x_{22}+x_{23}+x_{24}+x_{31}+x_{32}+x_{33}+x_{34}+x_{41}+x_{42}+x_{43}+x_{44} \\
 \text{subject to} & x_{11}+x_{12}+x_{13}+x_{14} \leq 1 \\
 & x_{21}+x_{22}+x_{23}+x_{24} \leq 1 \\
 & x_{31}+x_{32}+x_{33}+x_{34} \leq 1 \\
 & x_{41}+x_{42}+x_{43}+x_{44} \leq 1 \\
 & x_{11} \qquad \qquad +x_{21} \qquad \qquad +x_{31} \qquad \qquad +x_{41} \leq 1 \\
 & \quad x_{12} \qquad \qquad +x_{22} \qquad \qquad +x_{32} \qquad \qquad +x_{42} \leq 1 \\
 & \qquad x_{13} \qquad \qquad +x_{23} \qquad \qquad +x_{33} \qquad \qquad +x_{43} \leq 1 \\
 & \qquad \qquad x_{14} \qquad \qquad +x_{24} \qquad \qquad +x_{34} \qquad \qquad +x_{44} \leq 1 \\
 & \qquad \qquad \quad x_{13} \qquad \qquad +x_{24} \qquad \qquad \leq 1 \\
 & \qquad \qquad \quad x_{12} \qquad \qquad +x_{23} \qquad \qquad +x_{34} \leq 1 \\
 & \qquad \qquad \quad x_{11} \qquad \qquad +x_{22} \qquad \qquad +x_{33} \qquad \qquad +x_{44} \leq 1 \\
 & \qquad \qquad \qquad x_{21} \qquad \qquad +x_{32} \qquad \qquad +x_{43} \leq 1 \\
 & \qquad \qquad \qquad \qquad x_{31} \qquad \qquad +x_{42} \leq 1 \\
 & \qquad \qquad \quad x_{12} \qquad \qquad +x_{21} \qquad \qquad \leq 1 \\
 & \qquad \qquad \quad x_{13} \qquad \qquad +x_{22} \qquad \qquad +x_{31} \leq 1 \\
 & \qquad \qquad \quad x_{14} \qquad \qquad +x_{23} \qquad \qquad +x_{32} \qquad \qquad +x_{41} \leq 1 \\
 & \qquad \qquad \qquad \qquad x_{24} \qquad \qquad +x_{33} \qquad \qquad +x_{42} \leq 1 \\
 & \qquad \qquad \qquad \qquad \qquad x_{34} \qquad \qquad +x_{43} \leq 1 \\
 & x_{11}, x_{12}, x_{13}, x_{14}, x_{21}, x_{22}, x_{23}, x_{24}, x_{31}, x_{32}, x_{33}, x_{34}, x_{41}, x_{42}, x_{43}, x_{44} \in \{0,1\}
 \end{array}$$

$$LB = 3 \cdot UB = 4 \cdot z(LP) = 4 \cdot z(LP^+) = 4 \cdot z(MILP) = 4$$

EX 10 — selection with a tool set

$$\begin{array}{ll}
 \max & 2000x_1 + 1500x_2 + 1800x_3 + 1700x_4 + 800x_5 \\
 \text{subject to} & \\
 & x_1 \qquad \qquad \qquad y_1 + y_2 + y_3 + y_4 + y_5 + y_6 \leq 4 \\
 & x_1 \qquad \qquad \qquad -y_1 \leq 0 \\
 & x_1 \qquad \qquad \qquad \qquad -y_3 \leq 0 \\
 & x_1 \qquad \qquad \qquad \qquad \qquad -y_4 \leq 0 \\
 & x_1 \qquad \qquad \qquad \qquad \qquad \qquad -y_5 \leq 0 \\
 & \qquad x_2 \qquad \qquad -y_1 \leq 0 \\
 & \qquad x_2 \qquad \qquad -y_2 \leq 0 \\
 & \qquad x_2 \qquad \qquad \qquad \qquad -y_6 \leq 0 \\
 & \qquad \qquad x_3 \qquad -y_1 \leq 0 \\
 & \qquad \qquad x_3 \qquad -y_2 \leq 0 \\
 & \qquad \qquad x_3 \qquad \qquad -y_3 \leq 0 \\
 & \qquad \qquad x_3 \qquad \qquad \qquad -y_5 \leq 0 \\
 & \qquad \qquad \qquad x_4 \qquad -y_2 \leq 0 \\
 & \qquad \qquad \qquad x_4 \qquad \qquad -y_4 \leq 0 \\
 & \qquad \qquad \qquad x_4 \qquad \qquad -y_5 \leq 0 \\
 & \qquad \qquad \qquad \qquad x_5 \qquad -y_5 \leq 0 \\
 & \qquad \qquad \qquad \qquad x_5 \qquad \qquad -y_6 \leq 0 \\
 & x_1, \quad x_2, \quad x_3, \quad x_4, \quad x_5 \in \{0, 1\} \\
 & \qquad \qquad \qquad y_1, y_2, y_3, y_4, y_5, y_6 \in \{0, 1\}
 \end{array}$$

$$LB = 2000 \cdot UB = \frac{23000}{3} \cdot z(LP) = 5200 \cdot z(LP^+) = 5200 \cdot z(MILP) = 2500$$

EX 11 — three types of shoe with a threshold

$$\begin{array}{llllll}
 \max & 150x_1 + 120x_2 + 130x_3 & & & & \\
 \text{subject to} & 850x_1 + 600x_2 + 700x_3 & & & & \leq 120000 \\
 & 3x_1 + 2x_2 + \frac{5}{2}x_3 & & & & \leq 7000 \\
 & 20x_1 + 15x_2 + 20x_3 & & & & \leq 40000 \\
 & x_1 & & -100y_1 & & \geq 0 \\
 & & x_2 & & -200y_2 & \geq 0 \\
 & & & x_3 & & -150y_3 \geq 0 \\
 & x_1 & & -141y_1 & & \leq 0 \\
 & & x_2 & & -200y_2 & \leq 0 \\
 & & & x_3 & & -171y_3 \leq 0 \\
 & x_1, & x_2, & x_3 & & \in \mathbb{Z}_{\geq 0} \\
 & & & & y_1, & y_2, & y_3 \in \{0, 1\}
 \end{array}$$

$$LB = 24000 \cdot UB = 24000 \cdot z(LP) = 24000 \cdot z(LP^+) = 24000 \cdot z(MILP) = 24000$$

EX 12 — workloads as balanced as possible

$$\begin{array}{ll} \min & z \\ \text{subject to} & 2x_1 + 3x_2 + 6x_3 + 7x_4 - z \leq 0 \\ & -2x_1 - 3x_2 - 6x_3 - 7x_4 - z \leq -18 \\ & x_1, \quad x_2, \quad x_3, \quad x_4 \in \{0, 1\} \\ & z \geq 0 \end{array}$$

$$UB = 9 \cdot LB = 9 \cdot z(LP) = 9 \cdot z(LP^+) = 9 \cdot z(MILP) = 9$$

EX 13 — weekly shifts

$$\begin{array}{llllllll}
 \min & 450x_1 + 455x_2 + 475x_3 + 490x_4 + 490x_5 + 490x_6 + 480x_7 \\
 \text{subject to} & x_1 & & & +\frac{1}{2}x_4 & +x_5 & +x_6 & +x_7 & \geq & 10 \\
 & x_1 & +x_2 & & & +\frac{1}{2}x_5 & +x_6 & +x_7 & \geq & 8 \\
 & x_1 & +x_2 & +x_3 & & & +\frac{1}{2}x_6 & +x_7 & \geq & 12 \\
 & x_1 & +x_2 & +x_3 & +x_4 & & & +\frac{1}{2}x_7 & \geq & 9 \\
 & \frac{1}{2}x_1 & +x_2 & +x_3 & +x_4 & +x_5 & & & \geq & 9 \\
 & & \frac{1}{2}x_2 & +x_3 & +x_4 & +x_5 & +x_6 & & \geq & 7 \\
 & & & \frac{1}{2}x_3 & +x_4 & +x_5 & +x_6 & +x_7 & \geq & 8 \\
 & x_1, & x_2, & x_3, & x_4, & x_5, & x_6, & x_7 & \in & \mathbb{Z}_{\geq 0}
 \end{array}$$

$$UB = 8410 \cdot LB = 6300 \cdot z(LP) = \frac{115970}{17} \cdot z(LP^+) = \frac{115970}{17} \cdot z(MILP) = 7060$$

EX 14 — production and workforce over three months

$$\begin{array}{ll}
\min & 15x_1 + 15x_2 + 15x_3 + 3s_1 + 3s_2 + 1500y_1 + 1500y_2 + 1500y_3 + 100z_1 + 100z_2 + 100z_3 \\
\text{subject to} & x_1 - s_1 = 2500 \\
& x_2 + s_1 - s_2 = 5000 \\
& x_3 + s_2 = 7000 \\
& -4x_1 + 160y_1 \geq 0 \\
& -4x_2 + 160y_2 \geq 0 \\
& -4x_3 + 160y_3 \geq 0 \\
& y_1 - z_1 = 100 \\
& -y_1 + y_2 - z_2 = 0 \\
& -y_2 + y_3 - z_3 = 0 \\
& x_1, x_2, x_3 \geq 0 \\
& s_1, s_2 \geq 0 \\
& y_1, y_2, y_3 \in \mathbb{Z}_{\geq 0} \\
& z_1, z_2, z_3 \in \mathbb{Z}_{> 0}
\end{array}$$

$$UB = 825000 \cdot LB = 761250 \cdot z(LP) = 773500 \cdot z(LP^+) = 773500 \cdot z(MILP) = 774180$$

EX 15 — weekly timetable

$$\begin{array}{llllllll}
 \min & & x_{112} & & +x_{122}+x_{211} & & +x_{221} & \\
 \text{subject to} & x_{111} & & +x_{121} & & +x_{211} & & +x_{221} & = 2 \\
 & & x_{112} & & +x_{122} & & +x_{212} & & +x_{222} & = 2 \\
 & & & x_{111}+x_{112} & & & & & & \leq 1 \\
 & & & & x_{121}+x_{122} & & & & & \leq 1 \\
 & & & & & x_{211}+x_{212} & & & & \leq 1 \\
 & & & & & & x_{221}+x_{222} & & & \leq 1 \\
 & & & & & & & y_{11}+y_{12} & & \geq 2 \\
 & & & & & & & & y_{21}+y_{22} & \geq 2 \\
 & x_{111} & & & & & & -y_{11} & & \leq 0 \\
 & & x_{112} & & & & & & -y_{12} & \leq 0 \\
 & & & x_{121} & & & & -y_{11} & & \leq 0 \\
 & & & & x_{122} & & & -y_{12} & & \leq 0 \\
 & & & & & x_{211} & & -y_{21} & & \leq 0 \\
 & & & & & & x_{212} & & -y_{22} & \leq 0 \\
 & & & & & & & x_{221} & & -y_{21} & \leq 0 \\
 & & & & & & & & x_{222} & & -y_{22} & \leq 0 \\
 & -x_{111} & & -x_{121} & & & & & +y_{11} & & \leq 0 \\
 & & -x_{112} & & -x_{122} & & & & & +y_{12} & \leq 0 \\
 & & & & & -x_{211} & & -x_{221} & & & +y_{21} & \leq 0 \\
 & & & & & & -x_{212} & & -x_{222} & & & +y_{22} & \leq 0 \\
 & x_{111}, x_{112}, x_{121}, x_{122}, x_{211}, x_{212}, x_{221}, x_{222} & & & & & & & & & \in \{0,1\} \\
 & & & & & & & & y_{11}, y_{12}, y_{21}, y_{22} & & \in \{0,1\}
 \end{array}$$

$$UB = 2 \cdot LB = 0 \cdot z(LP) = 0 \cdot z(LP^+) = 2 \cdot z(MILP) = 2$$

The twenty-three problems

Twenty-three problems with a symbolic model

For each problem: the statement as it arrives, the symbolic model and the five numbers of the sandwich. The models are those of the notes, generated from the same source.

- LB and UB are the two sides of the sandwich built *by hand*: the heuristic solution on one side, the dual certificate on the other.
- $z(LP)$ is the relaxation without the bounds, $z(LP^+)$ the one with them, $z(MILP)$ the integer optimum.
- In a minimisation the heuristic gives UB and the dual LB ; in a maximisation the two roles swap.

Problem 7.1 — minimum-cost assignment

The statement

A company has to process $n \in \mathbb{Z}_{\geq 1}$ jobs on $k \in \mathbb{Z}_{\geq 1}$ machines. For each job $j \in \{1, 2, \dots, n\}$ and each machine $m \in \{1, 2, \dots, k\}$, the value $t_{jm} \in \mathbb{Q}_{>0}$ is the processing time in minutes and the value $c_{jm} \in \mathbb{Q}_{>0}$ is the cost in euros of processing job j on machine m . For each machine $m \in \{1, 2, \dots, k\}$, the value $a_m \in \mathbb{Q}_{>0}$ is the available processing time in minutes. Each machine handles one job at a time (no parallel processing). The company seeks a minimum-cost assignment of all jobs to the machines.

Model 7.1 — minimum-cost assignment

$$\min \sum_{j=1}^n \sum_{m=1}^k c_{jm} x_{jm}$$

$$\text{subject to } \sum_{m=1}^k x_{jm} = 1, \quad \forall j \in \{1, 2, \dots, n\},$$

$$\sum_{j=1}^n t_{jm} x_{jm} \leq a_m, \quad \forall m \in \{1, 2, \dots, k\},$$

$$x_{jm} \in \{0, 1\}, \quad \forall j \in \{1, 2, \dots, n\}, \quad \forall m \in \{1, 2, \dots, k\}.$$

$$UB = 11 \cdot LB = 10 \cdot z(LP) = \frac{53}{5} \cdot z(LP^+) = \frac{53}{5} \cdot z(MILP) = 11$$

Problem 7.2 — machines with fixed cost

The statement

A company needs to process $n \in \mathbb{Z}_{\geq 1}$ jobs and has $k \in \mathbb{Z}_{\geq 1}$ machines. For each job $j \in \{1, 2, \dots, n\}$ and each machine $m \in \{1, 2, \dots, k\}$, the value $t_{jm} \in \mathbb{Q}_{>0}$ is the processing time in minutes of job j on machine m . For each machine $m \in \{1, 2, \dots, k\}$, the value $a_m \in \mathbb{Q}_{>0}$ is the availability in minutes and the value $c_m \in \mathbb{Q}_{>0}$ is the cost in euros if the machine is used. Each machine handles one job at a time. The company wants to assign all jobs to the machines minimising the cost of the machines used.

Model 7.2 — machines with fixed cost

$$\begin{aligned} \min \quad & \sum_{m=1}^k c_m y_m \\ \text{subject to} \quad & \sum_{m=1}^k x_{jm} = 1, & \forall j \in \{1, 2, \dots, n\}, \\ & - \sum_{j=1}^n t_{jm} x_{jm} + a_m y_m \geq 0, & \forall m \in \{1, 2, \dots, k\}, \\ & x_{jm} \in \{0, 1\}, \quad \forall j \in \{1, 2, \dots, n\}, \quad \forall m \in \{1, 2, \dots, k\}, \\ & y_m \in \{0, 1\}, & \forall m \in \{1, 2, \dots, k\}. \end{aligned}$$

$$UB = 12 \quad \cdot \quad LB = \frac{25}{4} \quad \cdot \quad z(LP) = \frac{25}{4} \quad \cdot \quad z(LP^+) = \frac{1273}{200} \quad \cdot \quad z(MILP) = 12$$

Problem 7.3 — job selection

The statement

A company can execute $n \in \mathbb{Z}_{\geq 1}$ jobs and has $k \in \mathbb{Z}_{\geq 1}$ machines. For each job $j \in \{1, 2, \dots, n\}$, the value $t_j \in \mathbb{Q}_{>0}$ is the processing time in minutes (the same on every machine) and the value $r_j \in \mathbb{Q}_{>0}$ is the revenue in euros if the job is executed. For each machine $m \in \{1, 2, \dots, k\}$, the value $a_m \in \mathbb{Q}_{>0}$ is the availability in minutes and the value $c_m \in \mathbb{Q}_{>0}$ is the cost in euros if the machine is used. Each machine handles one job at a time. The company wants to choose which jobs to execute, and on which machines, to maximise the profit: revenues of the executed jobs minus costs of the machines used.

Model 7.3 — job selection

$$\begin{aligned} \max \quad & \sum_{j=1}^n \sum_{m=1}^k r_j x_{jm} - \sum_{m=1}^k c_m y_m \\ \text{subject to} \quad & \sum_{m=1}^k x_{jm} \leq 1, & \forall j \in \{1, 2, \dots, n\}, \\ & \sum_{j=1}^n t_j x_{jm} - a_m y_m \leq 0, & \forall m \in \{1, 2, \dots, k\}, \\ & x_{jm} \in \{0, 1\}, \quad \forall j \in \{1, 2, \dots, n\}, \quad \forall m \in \{1, 2, \dots, k\}, \\ & y_m \in \{0, 1\}, & \forall m \in \{1, 2, \dots, k\}. \end{aligned}$$

$$LB = 20 \cdot UB = 34 \cdot z(LP) = 34 \cdot z(LP^+) = \frac{680}{21} \cdot z(MILP) = 25$$

Problem 7.4 — parallel jobs

The statement

A company needs to execute $n \in \mathbb{Z}_{\geq 1}$ jobs using $k \in \mathbb{Z}_{\geq 1}$ machines. For each job $j \in \{1, 2, \dots, n\}$ and each machine $m \in \{1, 2, \dots, k\}$, the value $t_{jm} \in \mathbb{Q}_{>0}$ is the processing time in minutes of job j on machine m . Each machine $m \in \{1, 2, \dots, k\}$ can execute at most $p_m \in \mathbb{Z}_{\geq 1}$ jobs, and executes them in parallel: all the jobs assigned to the same machine start together. The company wants to assign the jobs minimising the sum of the processing times of the machines, where the processing time of a machine is the time of the longest job among those assigned to it.

Model 7.4 — parallel jobs

$$\begin{aligned} \min \quad & \sum_{m=1}^k y_m \\ \text{subject to} \quad & \sum_{m=1}^k x_{jm} = 1, & \forall j \in \{1, 2, \dots, n\}, \\ & \sum_{j=1}^n x_{jm} \leq p_m, & \forall m \in \{1, 2, \dots, k\}, \\ & -t_{jm} x_{jm} + y_m \geq 0, & \forall j \in \{1, 2, \dots, n\}, \forall m \in \{1, 2, \dots, k\}, \\ & x_{jm} \in \{0, 1\}, & \forall j \in \{1, 2, \dots, n\}, \forall m \in \{1, 2, \dots, k\}, \\ & y_m \geq 0, & \forall m \in \{1, 2, \dots, k\}. \end{aligned}$$

$$UB = 19 \cdot LB = 5 \cdot z(LP) = \frac{520}{49} \cdot z(LP^+) = \frac{520}{49} \cdot z(MILP) = 15$$

Problem 7.5 — classes with setup

The statement

A company has $n \in \mathbb{Z}_{\geq 1}$ jobs that can be executed on a machine with an available processing time of $a \in \mathbb{Q}_{>0}$ minutes. For each job $j \in \{1, 2, \dots, n\}$, the value $t_j \in \mathbb{Q}_{>0}$ is the processing time in minutes and the value $r_j \in \mathbb{Q}_{>0}$ is the revenue in euros if the job is executed. The jobs are partitioned into $q \in \mathbb{Z}_{\geq 2}$ classes: for each class $c \in \{1, 2, \dots, q\}$, the set $\mathcal{J}_c \subseteq \{1, 2, \dots, n\}$ contains the jobs of the class. If the machine executes jobs of a class c , a setup cost of $f_c \in \mathbb{Q}_{\geq 0}$ euros is incurred and a setup time of $s_c \in \mathbb{Q}_{\geq 0}$ minutes is needed to prepare the machine. The machine cannot execute jobs in parallel. The company wants to choose the jobs to execute maximising the profit.

Model 7.5 — classes with setup

$$\begin{aligned} \max \quad & \sum_{j=1}^n r_j x_j - \sum_{c=1}^q f_c y_c \\ \text{subject to} \quad & \sum_{j=1}^n t_j x_j + \sum_{c=1}^q s_c y_c \leq a, \\ & x_j - y_c \leq 0, \quad \forall c \in \{1, 2, \dots, q\}, \quad \forall j \in \mathcal{J}_c, \\ & x_j \in \{0, 1\}, \quad \forall j \in \{1, 2, \dots, n\}, \\ & y_c \in \{0, 1\}, \quad \forall c \in \{1, 2, \dots, q\}. \end{aligned}$$

$$LB = 9 \quad \cdot \quad UB = 100 \quad \cdot \quad z(LP) = \frac{425}{13} \quad \cdot \quad z(LP^+) = \frac{329}{13} \quad \cdot \quad z(MILP) = 21$$

Problem 7.6 — completion bonus

The statement

A company has $n \in \mathbb{Z}_{\geq 1}$ jobs that can be executed on a machine with an available processing time of $a \in \mathbb{Q}_{>0}$ minutes. For each job $j \in \{1, 2, \dots, n\}$, $t_j \in \mathbb{Q}_{>0}$ is the processing time in minutes and $r_j \in \mathbb{Q}_{>0}$ the revenue in euros if executed. The jobs are partitioned into $q \in \mathbb{Z}_{\geq 2}$ classes $\mathcal{J}_1, \dots, \mathcal{J}_q$. The availability of the machine is reduced by $u \in \mathbb{Q}_{>0}$ minutes *if and only if* the machine executes jobs of at least two different classes. For each class $c \in \{1, 2, \dots, q\}$, an extra revenue of $v_c \in \mathbb{Q}_{>0}$ euros is obtained *if and only if* all the jobs of the class are executed. The machine does not execute jobs in parallel. The company wants to choose the jobs to execute maximising the total revenue.

Model 7.6 — completion bonus

$$\begin{aligned} \max \quad & \sum_{j=1}^n r_j x_j + \sum_{c=1}^q v_c y_c \\ \text{subject to} \quad & x_j - y_c \geq 0, & \forall c \in \{1, 2, \dots, q\}, \forall j \in \mathcal{J}_c, \\ & x_j + x_i - z \leq 1, & \forall c, g \in \{1, 2, \dots, q\}, c < g, \forall j \in \mathcal{J}_c, \forall i \in \mathcal{J}_g, \\ & \sum_{j=1}^n t_j x_j + u z \leq a, \\ & x_j \in \{0, 1\}, & \forall j \in \{1, 2, \dots, n\}, \\ & y_c \in \{0, 1\}, & \forall c \in \{1, 2, \dots, q\}, \\ & z \in \{0, 1\}. \end{aligned}$$

$$LB = 32 \cdot UB = 150 \cdot z(LP) = \frac{5280}{113} \cdot z(LP^+) = \frac{5280}{113} \cdot z(MILP) = 42$$

Problem 7.7 — total tardiness

The statement

A company needs to process $n \in \mathbb{Z}_{\geq 1}$ jobs on a single machine. For each job $j \in \{1, 2, \dots, n\}$, $t_j \in \mathbb{Q}_{>0}$ is the processing time in minutes and $d_j \in \mathbb{Q}_{>0}$ the due date in minutes. The tardiness of job j is $\tau_j = \max\{0, \kappa_j - d_j\}$, where κ_j is the completion time. The machine processes one job at a time. The company wants a sequence of the jobs that minimises the total tardiness.

Model 7.7 — total tardiness

$$\min \sum_{j=1}^n \tau_j$$

$$\begin{aligned} \text{subject to } s_{ji} + s_{ij} &= 1, & \forall j, i \in \{1, 2, \dots, n\}, j < i, \\ -M s_{ji} - \kappa_j + \kappa_i &\geq t_i - M, & \forall j, i \in \{1, 2, \dots, n\}, j \neq i, \\ -\kappa_j + \tau_j &\geq -d_j, & \forall j \in \{1, 2, \dots, n\}, \\ \kappa_j &\geq t_j, & \forall j \in \{1, 2, \dots, n\}, \\ s_{ji} &\in \{0, 1\}, & \forall j, i \in \{1, 2, \dots, n\}, j \neq i, \\ \kappa_j &\geq 0, & \forall j \in \{1, 2, \dots, n\}, \\ \tau_j &\geq 0, & \forall j \in \{1, 2, \dots, n\}. \end{aligned}$$

$$UB = 12 \cdot LB = 2 \cdot z(LP) = 2 \cdot z(LP^+) = 2 \cdot z(MILP) = 11$$

Problem 8.1 — capacitated facility location

The statement

A company must serve $n \in \mathbb{Z}_{\geq 1}$ clients with its product and has identified $m \in \mathbb{Z}_{\geq 1}$ candidate locations where a facility can be installed. For each client $c \in \{1, 2, \dots, n\}$, the value $d_c \in \mathbb{Q}_{>0}$ is the demand, in liters of product, of that client. For each location $l \in \{1, 2, \dots, m\}$ and each client c , the value $t_{lc} \in \mathbb{Q}_{>0}$ is the cost, in euros, of transporting one liter of product from location l to client c . For each location l , the value $u_l \in \mathbb{Q}_{>0}$ is the capacity, in liters, of the facility installed there, and $i_l \in \mathbb{Q}_{>0}$ is the installation cost. The company wants to decide where to install facilities and how to serve the clients, at minimum total cost.

Model 8.1 — capacitated facility location

$$\begin{aligned} \min \quad & \sum_{l=1}^m i_l x_l + \sum_{l=1}^m \sum_{c=1}^n t_{lc} y_{lc} \\ \text{subject to} \quad & u_l x_l - \sum_{c=1}^n y_{lc} \geq 0, & \forall l \in \{1, 2, \dots, m\}, \\ & \sum_{l=1}^m y_{lc} = d_c, & \forall c \in \{1, 2, \dots, n\}, \\ & x_l \in \{0, 1\}, & \forall l \in \{1, 2, \dots, m\}, \\ & y_{lc} \geq 0, & \forall l \in \{1, 2, \dots, m\}, \forall c \in \{1, 2, \dots, n\}. \end{aligned}$$

$$UB = 439 \quad \cdot \quad LB = \frac{1581}{5} \quad \cdot \quad z(LP) = \frac{1581}{5} \quad \cdot \quad z(LP^+) = 317 \quad \cdot \quad z(MILP) = 365$$

Problem 8.2 — p-median

The statement

A company must choose at most $k \in \mathbb{Z}_{\geq 1}$ locations, among $m \in \mathbb{Z}_{\geq 1}$ candidates, where to open a service point, and assign each of the $n \in \mathbb{Z}_{\geq 1}$ clients to the most convenient open location. For each location $l \in \{1, 2, \dots, m\}$ and each client $c \in \{1, 2, \dots, n\}$, the value $d_{lc} \in \mathbb{Q}_{>0}$ is the distance (or service cost) between location l and client c . Every client must be assigned to exactly one open location. The company wants to decide which locations to open and how to assign clients, minimizing the sum of the distances between each client and the location it is assigned to.

Model 8.2 — p-median

$$\begin{aligned} \min \quad & \sum_{l=1}^m \sum_{c=1}^n d_{lc} y_{lc} \\ \text{subject to} \quad & \sum_{l=1}^m y_{lc} = 1, & \forall c \in \{1, 2, \dots, n\}, \\ & \sum_{l=1}^m x_l \leq k, \\ & x_l - y_{lc} \geq 0, & \forall l \in \{1, 2, \dots, m\}, \forall c \in \{1, 2, \dots, n\}, \\ & x_l \in \{0, 1\}, & \forall l \in \{1, 2, \dots, m\}, \\ & y_{lc} \in \{0, 1\}, & \forall l \in \{1, 2, \dots, m\}, \forall c \in \{1, 2, \dots, n\}. \end{aligned}$$

$$UB = 18 \cdot LB = 13 \cdot z(LP) = 15 \cdot z(LP^+) = 15 \cdot z(MILP) = 15$$

Problem 8.3 — coverage with interference

The statement

An operator must choose at most $k \in \mathbb{Z}_{\geq 1}$ locations, among $m \in \mathbb{Z}_{\geq 1}$ candidates, where to install an antenna, to serve $n \in \mathbb{Z}_{\geq 1}$ clients. For each location $l \in \{1, 2, \dots, m\}$ and each client $c \in \{1, 2, \dots, n\}$, the value $s_{lc} \in \mathbb{Q}_{\geq 0}$ is the signal strength received by client c if location l is installed. A client c is *covered* if and only if the total signal strength received from all installed locations is at least the threshold $t \in \mathbb{Q}_{>0}$ *and* at most one installed location generates for that client a signal greater than or equal to the interference limit $b \in \mathbb{Q}_{>0}$. For each client c , the value $p_c \in \mathbb{Q}_{>0}$ is the profit if the client is covered. The operator wants to decide which locations to install to maximize the total profit.

Model 8.3 — coverage with interference

$$\begin{aligned} & \max \quad \sum_{c=1}^n p_c y_c \\ \text{subject to} \quad & - \sum_{l=1}^m s_{lc} x_l + t y_c \leq 0, \quad \forall c \in \{1, 2, \dots, n\}, \\ & \sum_{l \in \mathcal{L}_c} x_l + (m-1) y_c \leq m, \quad \forall c \in \{1, 2, \dots, n\}, \\ & \sum_{l=1}^m x_l \leq k, \\ & x_l \in \{0, 1\}, \quad \forall l \in \{1, 2, \dots, m\}, \\ & y_c \in \{0, 1\}, \quad \forall c \in \{1, 2, \dots, n\}. \end{aligned}$$

$$LB = 25 \quad \cdot \quad UB = \frac{225}{2} \quad \cdot \quad z(LP) = \frac{41925}{646} \quad \cdot \quad z(LP^+) = \frac{125}{2} \quad \cdot \quad z(MILP) = 45$$

Problem 8.4 — hub with maximum cost

The statement

In a computer network there are $n \in \mathbb{Z}_{\geq 1}$ terminals, each of which must be connected to exactly one hub. There are $m \in \mathbb{Z}_{\geq 1}$ hubs available, and each hub can serve at most $k \in \mathbb{Z}_{\geq 1}$ terminals. A hub must be activated before it can serve any terminals: for each hub $j \in \{1, 2, \dots, m\}$, the value $f_j \in \mathbb{Q}_{\geq 0}$ is the activation cost. For each terminal $i \in \{1, 2, \dots, n\}$ and each hub j , the value $c_{ij} \in \mathbb{Q}_{\geq 0}$ is the cost of connecting terminal i to hub j . The objective is to decide how to assign terminals to hubs to minimize the sum of the activation costs of the hubs used and the maximum connection cost of each hub, subject to the constraints described.

Model 8.4 — hub with maximum cost

$$\begin{aligned} \min \quad & \sum_{j=1}^m f_j y_j + \sum_{j=1}^m z_j \\ \text{subject to} \quad & \sum_{j=1}^m x_{ij} = 1, & \forall i \in \{1, 2, \dots, n\}, \\ & -\sum_{i=1}^n x_{ij} + k y_j \geq 0, & \forall j \in \{1, 2, \dots, m\}, \\ & -c_{ij} x_{ij} + z_j \geq 0, & \forall i \in \{1, 2, \dots, n\}, \forall j \in \{1, 2, \dots, m\}, \\ & x_{ij} \in \{0, 1\}, & \forall i \in \{1, 2, \dots, n\}, \forall j \in \{1, 2, \dots, m\}, \\ & y_j \in \{0, 1\}, & \forall j \in \{1, 2, \dots, m\}, \\ & z_j \geq 0, & \forall j \in \{1, 2, \dots, m\}. \end{aligned}$$

$$UB = 20 \cdot LB = \frac{15}{2} \cdot z(LP) = \frac{25}{2} \cdot z(LP^+) = \frac{1015}{78} \cdot z(MILP) = 19$$

Problem 9.1 — lot sizing with a fixed cost

The statement

A company must plan the production of a single product over $n \in \mathbb{Z}_{\geq 1}$ periods. For every period $t \in \{1, 2, \dots, n\}$, the value $d_t \in \mathbb{Q}_{\geq 0}$ is the demand to be met, $p_t \in \mathbb{Q}_{>0}$ the cost of producing one unit and $q_t \in \mathbb{Q}_{\geq 0}$ the fixed cost to be paid if production is started in that period. For every period $t \in \{1, 2, \dots, n-1\}$, the value $h_t \in \mathbb{Q}_{\geq 0}$ is the cost of keeping one unit in stock from the end of period t to the start of the next one. The initial inventory is $r_0 \in \mathbb{Q}_{\geq 0}$ and the one required at the end of the horizon is $r_n \in \mathbb{Q}_{\geq 0}$. The company wants to decide how much to produce in every period so as to meet all the demand at minimum total cost.

Model 9.1 — lot sizing with a fixed cost

$$\begin{aligned} \min \quad & \sum_{t=1}^n p_t x_t + \sum_{t=1}^n q_t y_t + \sum_{t=1}^{n-1} h_t s_t \\ \text{subject to} \quad & x_1 - s_1 = d_1 - r_0, \\ & x_t + s_{t-1} - s_t = d_t, \quad \forall t \in \{2, 3, \dots, n-1\}, \\ & x_n + s_{n-1} = d_n + r_n, \\ & -x_t + M_t y_t \geq 0, \quad \forall t \in \{1, 2, \dots, n\}, \\ & x_t \geq 0, \quad \forall t \in \{1, 2, \dots, n\}, \\ & s_t \geq 0, \quad \forall t \in \{1, 2, \dots, n-1\}, \\ & y_t \in \{0, 1\}, \quad \forall t \in \{1, 2, \dots, n\}. \end{aligned}$$

$$UB = 420 \cdot LB = 270 \cdot z(LP) = \frac{3890}{11} \cdot z(LP^+) = \frac{3890}{11} \cdot z(MILP) = 390$$

Problem 9.2 — with the hirings

The statement

A shoe factory must plan its production over $n \in \mathbb{Z}_{\geq 1}$ months. For every month $t \in \{1, 2, \dots, n\}$, the value $d_t \in \mathbb{Z}_{\geq 0}$ is the demand in pairs and $p_t \in \mathbb{Q}_{>0}$ the cost of the raw materials for one pair. For every month $t \in \{1, 2, \dots, n-1\}$, the value $h_t \in \mathbb{Q}_{\geq 0}$ is the cost of keeping one pair in stock at the end of the month. At the start of the horizon the company has $m_0 \in \mathbb{Z}_{\geq 0}$ workers; every worker works $r \in \mathbb{Q}_{>0}$ hours a month and costs $w \in \mathbb{Q}_{>0}$ euros a month, and producing one pair requires $g \in \mathbb{Q}_{>0}$ labour hours. At the start of every month workers may be hired, at a cost of $u \in \mathbb{Q}_{\geq 0}$ euros each; nobody is laid off. The company wants to decide how much to produce and how many workers to employ in every month, at minimum total cost.

Model 9.2 — with the hirings

$$\begin{aligned} \min \quad & \sum_{t=1}^n p_t x_t + \sum_{t=1}^{n-1} h_t s_t + \sum_{t=1}^n (u + w(n - t + 1)) z_t \\ \text{subject to} \quad & x_1 - s_1 = d_1, \\ & x_t + s_{t-1} - s_t = d_t, \quad \forall t \in \{2, 3, \dots, n-1\}, \\ & x_n + s_{n-1} = d_n, \\ & -g x_t + r \sum_{j=1}^t z_j \geq -r m_0, \quad \forall t \in \{1, 2, \dots, n\}, \\ & x_t \in \mathbb{Z}_{\geq 0}, \quad \forall t \in \{1, 2, \dots, n\}, \\ & s_t \in \mathbb{Z}_{\geq 0}, \quad \forall t \in \{1, 2, \dots, n-1\}, \\ & z_t \in \mathbb{Z}_{\geq 0}, \quad \forall t \in \{1, 2, \dots, n\}. \end{aligned}$$

$$UB = 18200 \cdot LB = 13500 \cdot z(LP) = 15960 \cdot z(LP^+) = 15960 \cdot z(MILP) = 16660$$

Problem 9.3 — vehicles with minimum lot and bonus

The statement

A company produces $s \in \mathbb{Z}_{\geq 1}$ types of vehicle using $k \in \mathbb{Z}_{\geq 1}$ resources. For every resource $i \in \{1, 2, \dots, k\}$ and every type $j \in \{1, 2, \dots, s\}$, the value $a_{ij} \in \mathbb{Q}_{\geq 0}$ is the amount of resource i needed for one unit of type j , and $b_i \in \mathbb{Q}_{>0}$ is the availability of resource i . For every type j , the value $\bar{p}_j \in \mathbb{Q}_{>0}$ is the profit of one unit and $\bar{q}_j \in \mathbb{Z}_{\geq 1}$ the minimum quantity to be produced if that type is chosen. If the production includes at least two different types, the company collects a bonus $\bar{r} \in \mathbb{Q}_{>0}$ (a grant for diversification). The company wants to maximise the total profit.

Model 9.3 — vehicles with minimum lot and bonus

$$\begin{aligned} \max \quad & \sum_{j=1}^s \bar{p}_j x_j + \bar{r} z \\ \text{subject to} \quad & \sum_{j=1}^s a_{ij} x_j \leq b_i, & \forall i \in \{1, 2, \dots, k\}, \\ & x_j - \bar{q}_j y_j \geq 0, & \forall j \in \{1, 2, \dots, s\}, \\ & x_j - M_j y_j \leq 0, & \forall j \in \{1, 2, \dots, s\}, \\ & -\sum_{j=1}^s y_j + 2z \leq 0, \\ & x_j \in \mathbb{Z}_{\geq 0}, & \forall j \in \{1, 2, \dots, s\}, \\ & y_j \in \{0, 1\}, & \forall j \in \{1, 2, \dots, s\}, \\ & z \in \{0, 1\}. \end{aligned}$$

$$LB = 9200 \cdot UB = 11250 \cdot z(LP) = \frac{20625}{2} \cdot z(LP^+) = 9750 \cdot z(MILP) = 9700$$

Problem 10.1 — prizes with two modes

The statement

A loyalty programme offers $s \in \mathbb{Z}_{\geq 1}$ prizes and a customer has $p \in \mathbb{Q}_{>0}$ points. Every prize $i \in \{1, 2, \dots, s\}$ can be obtained in two alternative ways: with points only, spending $a_i \in \mathbb{Q}_{>0}$ of them; or spending only $b_i \in \mathbb{Q}_{>0}$ points (with $b_i < a_i$) and adding a cash contribution of $c_i \in \mathbb{Q}_{\geq 0}$ euros. Every prize can be obtained at most once, and in one mode only. Every prize carries a preference value $d_i \in \mathbb{Q}_{>0}$, and the customer wants an overall preference of at least $\ell \in \mathbb{Q}_{>0}$. The customer wants to minimise the money spent.

Model 10.1 — prizes with two modes

$$\begin{aligned} & \min \sum_{i=1}^s c_i y_i \\ & \text{subject to} \quad x_i + y_i \leq 1, \quad \forall i \in \{1, 2, \dots, s\}, \\ & \quad \sum_{i=1}^s (a_i x_i + b_i y_i) \leq p, \\ & \quad \sum_{i=1}^s d_i (x_i + y_i) \geq \ell, \\ & \quad x_i, y_i \in \{0, 1\}, \quad \forall i \in \{1, 2, \dots, s\}. \end{aligned}$$

$$UB = 8 \cdot LB = 3 \cdot z(LP) = 3 \cdot z(LP^+) = 3 \cdot z(MILP) = 5$$

Problem 10.2 — combinatorial auction

The statement

An auctioneer has a set $S = \{1, 2, \dots, n\}$ of $n \in \mathbb{Z}_{\geq 1}$ items to sell and has received $r \in \mathbb{Z}_{\geq 1}$ bids. For every bid $j \in \{1, 2, \dots, r\}$, the set $B_j \subseteq S$ is the subset of items requested and $p_j \in \mathbb{Q}_{>0}$ the profit in euros if the bid is accepted. Each item can be sold at most once, and a bid can be accepted only if all its items are available. The auctioneer wants to select a set of bids of maximum total profit.

Model 10.2 — combinatorial auction

$$\begin{aligned} & \max \quad \sum_{j=1}^r p_j x_j \\ & \text{subject to} \quad \sum_{j: i \in B_j} x_j \leq 1, & \forall i \in S, \\ & \quad \quad \quad x_j \in \{0, 1\}, \quad \forall j \in \{1, 2, \dots, r\}. \end{aligned}$$

$$LB = 21 \cdot UB = 23 \cdot z(LP) = 22 \cdot z(LP^+) = 22 \cdot z(MILP) = 22$$

Problem 10.3 — diet with variety

The statement

A nutritionist must compose a monthly diet choosing among $s \in \mathbb{Z}_{\geq 1}$ foods and controlling $r \in \mathbb{Z}_{\geq 1}$ nutrients. For every food $i \in \{1, 2, \dots, s\}$, the value $w_i \in \mathbb{Q}_{>0}$ is the cost of one kilo; for every food i and every nutrient $j \in \{1, 2, \dots, r\}$, the value $g_{ij} \in \mathbb{Q}_{\geq 0}$ is the amount of nutrient j contained in one kilo of food i . For every nutrient j , the monthly intake must lie between $a_j \in \mathbb{Q}_{>0}$ and $b_j \in \mathbb{Q}_{>0}$. If a food enters the diet, at least $c_i \in \mathbb{Q}_{>0}$ kilos of it must be consumed and at most $d_i \in \mathbb{Q}_{>0}$. The diet must include at least $t \in \mathbb{Z}_{\geq 1}$ different foods. The goal is the diet of minimum cost.

Model 10.3 — diet with variety

$$\begin{aligned} \min \quad & \sum_{i=1}^s w_i x_i \\ \text{subject to} \quad & \sum_{i=1}^s g_{ij} x_i \geq a_j, & \forall j \in \{1, 2, \dots, r\}, \\ & \sum_{i=1}^s g_{ij} x_i \leq b_j, & \forall j \in \{1, 2, \dots, r\}, \\ & x_i - c_i y_i \geq 0, & \forall i \in \{1, 2, \dots, s\}, \\ & x_i - d_i y_i \leq 0, & \forall i \in \{1, 2, \dots, s\}, \\ & \sum_{i=1}^s y_i \geq t, \\ & x_i \geq 0, & \forall i \in \{1, 2, \dots, s\}, \\ & y_i \in \{0, 1\}, & \forall i \in \{1, 2, \dots, s\}. \end{aligned}$$

$$UB = \frac{39}{4} \cdot LB = 9 \cdot z(LP) = \frac{46}{5} \cdot z(LP^+) = \frac{48}{5} \cdot z(MILP) = \frac{48}{5}$$

Problem 10.4 — trees and lights

The statement

The city administration wants to decorate $q \in \mathbb{Z}_{\geq 1}$ trees for the holidays. Every tree can be decorated according to one of the $n \in \mathbb{Z}_{\geq 1}$ possible configurations, each characterised by lights of $m \in \mathbb{Z}_{\geq 1}$ different colours. For every configuration $c \in \{1, 2, \dots, n\}$ and every colour $l \in \{1, 2, \dots, m\}$, the value $u_{cl} \in \mathbb{Z}_{\geq 0}$ is the number of lights of colour l required by configuration c , and $i_c \in \mathbb{Q}_{>0}$ is the installation cost of a tree decorated according to it. All the lights must be bought on the market, where they are sold in boxes of $k \in \mathbb{Z}_{\geq 1}$ types: for every type $b \in \{1, 2, \dots, k\}$ and every colour l , the value $v_{bl} \in \mathbb{Z}_{\geq 0}$ is the number of lights of colour l contained in a box of type b , and $p_b \in \mathbb{Q}_{\geq 0}$ is its cost. To ensure a pleasant visual variety at least $f \in \mathbb{Z}_{\geq 1}$ different configurations must be used. The goal is to decorate the trees at minimum total cost.

Model 10.4 — trees and lights

$$\begin{aligned} \min \quad & \sum_{c=1}^n i_c x_c + \sum_{b=1}^k p_b y_b \\ \text{subject to} \quad & \sum_{c=1}^n x_c = q, \\ & \sum_{b=1}^k v_{bl} y_b - \sum_{c=1}^n u_{cl} x_c \geq 0, \quad \forall l \in \{1, 2, \dots, m\}, \\ & \sum_{c=1}^n z_c \geq f, \\ & x_c - z_c \geq 0, \quad \forall c \in \{1, 2, \dots, n\}, \\ & x_c \in \mathbb{Z}_{\geq 0}, \quad \forall c \in \{1, 2, \dots, n\}, \\ & y_b \in \mathbb{Z}_{\geq 0}, \quad \forall b \in \{1, 2, \dots, k\}, \\ & z_c \in \{0, 1\}, \quad \forall c \in \{1, 2, \dots, n\}. \end{aligned}$$

$$UB = 3121 \cdot LB = 2140 \cdot z(LP) = 2140 \cdot z(LP^+) = 2141 \cdot z(MILP) = 2141$$

Problem 10.5 — shipments in boxes

The statement

A company needs to organise the shipments from its $n \in \mathbb{Z}_{\geq 1}$ production plants to its $m \in \mathbb{Z}_{\geq 1}$ customers. The company produces $k \in \mathbb{Z}_{\geq 1}$ product types. Products are shipped in boxes: every box travels directly from one plant to one customer and has maximum capacity $w \in \mathbb{Z}_{\geq 1}$, expressed as the maximum number of product units it can carry. For every product $p \in \{1, 2, \dots, k\}$ and every customer $c \in \{1, 2, \dots, m\}$, the value $d_{pc} \in \mathbb{Z}_{\geq 0}$ is the number of units ordered. For every product p and every plant $s \in \{1, 2, \dots, n\}$, the value $a_{ps} \in \mathbb{Z}_{\geq 0}$ is the number of units available. The company wants a shipment plan that minimises the total number of boxes used.

Model 10.5 — shipments in boxes

$$\begin{aligned} \min \quad & \sum_{s=1}^n \sum_{c=1}^m y_{sc} \\ \text{subject to} \quad & \sum_{s=1}^n x_{psc} = d_{pc}, & \forall p \in P, \forall c \in C, \\ & \sum_{c=1}^m x_{psc} \leq a_{ps}, & \forall p \in P, \forall s \in S, \\ & - \sum_{p=1}^k x_{psc} + w y_{sc} \geq 0, & \forall s \in S, \forall c \in C, \\ & x_{psc} \in \mathbb{Z}_{\geq 0}, & \forall p \in P, \forall s \in S, \forall c \in C, \\ & y_{sc} \in \mathbb{Z}_{\geq 0}, & \forall s \in S, \forall c \in C. \end{aligned}$$

$$UB = 2 \cdot LB = 2 \cdot z(LP) = \frac{11}{10} \cdot z(LP^+) = \frac{11}{10} \cdot z(MILP) = 2$$

Problem 10.6 — summer camps

The statement

A company runs $r \in \mathbb{Z}_{\geq 1}$ summer camps to host children during the holidays. For every camp $j \in \{1, 2, \dots, r\}$, the value $d_j \in \mathbb{Z}_{\geq 1}$ is the maximum number of children that can be hosted. The company has received requests from children of $s \in \mathbb{Z}_{\geq 1}$ different nationalities: for every nationality $i \in \{1, 2, \dots, s\}$ there are $f_i \in \mathbb{Z}_{\geq 0}$ girls and $g_i \in \mathbb{Z}_{\geq 0}$ boys. In every camp the number of girls must be greater than or equal to the number of boys, and the number of children of nationality $c \in \{1, 2, \dots, s\}$ must be greater than or equal to that of every other nationality. The company wants to maximise the total number of children accepted.

Model 10.6 — summer camps

$$\begin{aligned} \max \quad & \sum_{i=1}^s \sum_{j=1}^r (x_{ij} + y_{ij}) \\ \text{subject to} \quad & \sum_{j=1}^r x_{ij} \leq f_i, & \forall i \in \{1, 2, \dots, s\}, \\ & \sum_{j=1}^r y_{ij} \leq g_i, & \forall i \in \{1, 2, \dots, s\}, \\ & \sum_{i=1}^s (x_{ij} + y_{ij}) \leq d_j, & \forall j \in \{1, 2, \dots, r\}, \\ & \sum_{i=1}^s (x_{ij} - y_{ij}) \geq 0, & \forall j \in \{1, 2, \dots, r\}, \\ & x_{cj} + y_{cj} - x_{ij} - y_{ij} \geq 0, & \forall i \in \{1, 2, \dots, s\}, i \neq c, \forall j \in \{1, 2, \dots, r\}, \\ & x_{ij} \in \mathbb{Z}_{\geq 0}, & \forall i \in \{1, 2, \dots, s\}, \forall j \in \{1, 2, \dots, r\}, \\ & y_{ij} \in \mathbb{Z}_{\geq 0}, & \forall i \in \{1, 2, \dots, s\}, \forall j \in \{1, 2, \dots, r\}. \end{aligned}$$

$$LB = 15 \cdot UB = 23 \cdot z(LP) = 23 \cdot z(LP^+) = 23 \cdot z(MILP) = 23$$

Problem 10.7 — antitrust split

The statement

A company has $s \in \mathbb{Z}_{\geq 1}$ branches and sells $r \in \mathbb{Z}_{\geq 1}$ products. For every branch $i \in \{1, 2, \dots, s\}$ and every product $j \in \{1, 2, \dots, r\}$, the value $v_{ij} \in \mathbb{Q}_{\geq 0}$ is the revenue in millions of euros generated by that branch with that product. Because of a new antitrust regulation the company must split into two smaller companies; every branch is indivisible and must be assigned to exactly one of the two. The company wants to partition the branches minimising the maximum difference, over all the products, between the revenues of the two new companies.

Model 10.7 — antitrust split

$$\begin{aligned} & \min \quad z \\ \text{subject to} \quad & z - 2 \sum_{i=1}^s v_{ij} x_i + T_j \geq 0, \quad \forall j \in \{1, 2, \dots, r\}, \\ & z + 2 \sum_{i=1}^s v_{ij} x_i - T_j \geq 0, \quad \forall j \in \{1, 2, \dots, r\}, \\ & x_i \in \{0, 1\}, \quad \forall i \in \{1, 2, \dots, s\}, \\ & z \begin{matrix} \geq \\ \leq \end{matrix} 0. \end{aligned}$$

$$UB = 6 \cdot LB = 2 \cdot z(LP) = 0 \cdot z(LP^+) = 0 \cdot z(MILP) = 4$$

Problem 10.8 — songs on CDs

The statement

A music production company publishes a collection of $n \in \mathbb{Z}_{\geq 1}$ songs. For every song $i \in \{1, 2, \dots, n\}$, the value $d_i \in \mathbb{Q}_{\geq 1}$ is the duration in minutes. The collection must be distributed over $m \in \mathbb{Z}_{\geq 1}$ compact discs; every song must be assigned to exactly one CD, and every CD $j \in \{1, 2, \dots, m\}$ must contain at least $w_j \in \mathbb{Z}_{\geq 1}$ songs. The company wants to minimise the difference between the total duration of the longest CD and that of the shortest one.

Model 10.8 — songs on CDs

$$\begin{aligned} & \min \quad y - z \\ \text{subject to} \quad & \sum_{j=1}^m x_{ij} = 1, & \forall i \in \{1, 2, \dots, n\}, \\ & \sum_{i=1}^n x_{ij} \geq w_j, & \forall j \in \{1, 2, \dots, m\}, \\ & -\sum_{i=1}^n d_i x_{ij} + y \geq 0, & \forall j \in \{1, 2, \dots, m\}, \\ & \sum_{i=1}^n d_i x_{ij} - z \geq 0, & \forall j \in \{1, 2, \dots, m\}, \\ & x_{ij} \in \{0, 1\}, \quad \forall i \in \{1, 2, \dots, n\}, \quad \forall j \in \{1, 2, \dots, m\}, \\ & y, z \geq 0. \end{aligned}$$

$$UB = 1 \cdot LB = 1 \cdot z(LP) = 0 \cdot z(LP^+) = 0 \cdot z(MILP) = 1$$

Problem 10.9 — books on shelves

The statement

A library needs to arrange $n \in \mathbb{Z}_{\geq 1}$ books on shelves. Every book $b \in \{1, 2, \dots, n\}$ has width $w_b \in \mathbb{Z}_{\geq 1}$ and height $h_b \in \mathbb{Z}_{\geq 1}$. There are $m \in \mathbb{Z}_{\geq 1}$ shelves available, each of maximum width $c \in \mathbb{Q}_{>0}$. Every book must be assigned to exactly one shelf, and the total width of the books on a shelf cannot exceed c . The height of a shelf is that of the tallest book placed on it. The library wants to minimise the sum of the heights of the shelves.

Model 10.9 — books on shelves

$$\begin{aligned} & \min \sum_{s=1}^m y_s \\ \text{subject to } & \sum_{s=1}^m x_{bs} = 1, & \forall b \in \{1, 2, \dots, n\}, \\ & \sum_{b=1}^n w_b x_{bs} \leq c, & \forall s \in \{1, 2, \dots, m\}, \\ & -h_b x_{bs} + y_s \geq 0, & \forall b \in \{1, 2, \dots, n\}, \forall s \in \{1, 2, \dots, m\}, \\ & x_{bs} \in \{0, 1\}, & \forall b \in \{1, 2, \dots, n\}, \forall s \in \{1, 2, \dots, m\}, \\ & y_s \geq 0, & \forall s \in \{1, 2, \dots, m\}. \end{aligned}$$

$$UB = 15 \cdot LB = 12 \cdot z(LP) = 8 \cdot z(LP^+) = 8 \cdot z(MILP) = 15$$

The forty problems to model

Forty problems to model

Here there are only the statements: twenty with explicit numerical data and twenty in symbolic form. The model is to be written, not read — these are the problems to practise on, and the solutions are reserved for instructors.

N1 — the projects to fund

The statement

A department chooses which projects to fund out of six. The costs are 40, 25, 30, 15, 50 and 20 thousand euros; the expected benefits are 9, 5, 7, 3, 11 and 4. The budget is 100. Projects 2 and 5 use the same laboratory and cannot both be funded; project 6 only makes sense if project 1 is funded too. The maximum benefit is wanted.

What is asked

Write the model and say how many constraints it has as a function of the data. What is the optimum of the instance?

N2 — the morning deliveries

The statement

Four deliveries must be given to three couriers. The table gives, for every courier and every delivery, the time needed in minutes:

	delivery 1	delivery 2	delivery 3	delivery 4
courier 1	20	35	25	40
courier 2	25	20	30	35
courier 3	30	25	20	30

Every delivery goes to one courier only; every courier works at most 60 minutes. The minimum total time is wanted.

What is asked

Write the model. What changes if one wants to minimise the time of the busiest courier instead?

N3 — the minimum coverage

The statement

A city has five districts and four possible antenna sites. Site 1 covers districts $\{1, 2\}$, site 2 covers $\{2, 3, 4\}$, site 3 covers $\{1, 4, 5\}$ and site 4 covers $\{3, 5\}$. Every antenna costs the same. All districts must be covered with the smallest number of antennas.

What is asked

Write the model and solve it. Then write the dual of the LP relaxation and build a feasible solution of it by hand: which bound does it give?

N4 — how many boxes are needed

The statement

Four parcels weigh 6, 5, 4 and 3 kilos. Every box carries at most 10 kilos and a parcel is not split. Three boxes are available. The minimum number of boxes is wanted.

What is asked

Write the model. Give a lower bound without solving the MILP, and say whether it is reached.

N5 — the integer product mix

The statement

A workshop makes two articles. One piece of the first takes 2 hours in department A, 1 in B and 3 in C; one piece of the second takes 1, 3 and 2. The departments have 20, 24 and 30 hours available. The unit margins are 7 and 6 euros, and pieces are sold whole only. The maximum margin is wanted.

What is asked

Write the model. How much is the linear relaxation worth, and what does integrality cost on this instance?

N6 — the counter clerks

The statement

A counter is open seven days. The clerks needed are, from Monday to Sunday, 4, 3, 4, 5, 6, 5 and 3. Every clerk hired works five consecutive days and then rests two; the cycle may start on any day of the week. The smallest number of clerks is wanted.

What is asked

Write the model. How many constraints does it have, and how many variables?

N7 — how much to buy from whom

The statement

100 units of a component are needed. Three suppliers sell it at 9, 7 and 8 euros a unit, but a supplier that receives an order also charges a fixed 50, 120 and 90 euros. The first supplier cannot exceed 40 units, the second 70, the third 60. The minimum spend is wanted.

What is asked

Write the model. What is the smallest M that can be used for each supplier, and why is it valid?

N8 — the worst tardiness

The statement

Four jobs must run one after the other on a single machine. The durations are 4, 2, 6 and 3 hours; the due dates are 8, 5, 14 and 10 hours from the start. The machine starts at time 0 and never stops. One wants to minimise the worst tardiness, that is the maximum between zero and the difference between completion and due date.

What is asked

Write the model with sequencing variables. What changes if one wants the sum of the tardinesses instead?

N9 — where to open

The statement

Three customers ask for 12, 8 and 15 pallets. Three warehouses can be opened, with capacities 20, 25 and 18 pallets and fixed opening costs 100, 90 and 80. The table gives the transport cost per pallet from each warehouse to each customer:

	customer 1	customer 2	customer 3
warehouse 1	2	4	5
warehouse 2	3	2	4
warehouse 3	5	3	2

The minimum total cost is wanted.

What is asked

Write the model. What is the link between opening and transport, and in which direction does the constraint impose it?

N10 — the balanced groups

The statement

Four students have average marks 28, 24, 30 and 23. They must be split into two groups of two. One wants the group with the highest average and the one with the lowest average to be as close as possible.

What is asked

Write the model with the difference between maximum and minimum. What is the value of the LP relaxation, and why?

N11 — the evening programme

The statement

Five films last 90, 120, 100, 140 and 110 minutes. Two screens are free for 240 minutes each. Every film is shown at most once, and in one screen only. The expected takings are 500, 700, 450, 800 and 600 euros. Films 2 and 4 have the same audience and must not be programmed in the same screen. The maximum takings are wanted.

What is asked

Write the model. Which link does the constraint on films 2 and 4 impose, and what changes if it is written once per screen instead of once in total?

N12 — the weekly ration

The statement

Four foods cost 2, 3, 1 and 4 euros a kilo. One kilo of each provides

	food 1	food 2	food 3	food 4
protein (g)	20	35	10	40
iron (mg)	3	2	5	4

At least 120 grams of protein and 25 milligrams of iron are needed. A food, if it enters the ration, enters for at least three kilos, and never for more than ten. The minimum spend is wanted.

What is asked

Write the model. What happens to the solution if the three-kilo constraint is dropped?

N13 — the quantity discount

The statement

A firm buys up to 200 units of a material. The price is 10 euros a unit for the first 50, 8 for those between 51 and 120, 7 beyond 120. The discount applies only to the units in the bracket, not to all of them. At least 90 units are needed, and the store holds at most 180. The minimum spend is wanted.

What is asked

Write the model with one variable per bracket and the adjacency constraints that prevent using a bracket before the previous one is full.

N14 — the cameras at the junctions

The statement

A road network joins six junctions with the stretches $\{1, 2\}$, $\{1, 3\}$, $\{2, 3\}$, $\{2, 4\}$, $\{3, 5\}$, $\{4, 5\}$, $\{4, 6\}$ and $\{5, 6\}$. A camera at a junction watches every stretch meeting there. All stretches must be watched with the smallest number of cameras.

What is asked

Write the model. Give a lower bound by looking at pairwise disjoint stretches.

N15 — set-ups and stocks

The statement

A line makes one article for three months. The demands are 100, 140 and 80 pieces. Making a piece costs 5 euros; starting production in a month costs 300 euros whatever the quantity; holding a piece at the end of a month costs 1 euro. The line makes at most 200 pieces a month. Stock starts and ends empty. The minimum cost is wanted.

What is asked

Write the model. What is the smallest valid M for each month?

N16 — the compatible load

The statement

Three containers are available: the first carries 20 tonnes and costs 100 euros, the second 25 tonnes and costs 120 euros, the third 15 tonnes and costs 80 euros; only the ones actually used are paid for. Four loads weigh 10, 8, 12 and 6 tonnes, and each must be shipped whole in a single container. Loads 1 and 3 are incompatible and do not travel together; load 4 requires a refrigerated container, and only the second one is. The minimum cost is wanted.

What is asked

Write the model. How many families of constraints are needed, and what does each one say?

N17 — the building site

The statement

Six activities last 3, 2, 4, 1, 5 and 2 days. Two crews work in parallel, and an activity occupies a single crew for its whole duration, without interruption. Every crew carries out its activities one after the other. One wants to finish as early as possible.

What is asked

Write the model with the assignment to the crews and the finishing time. Which lower bound can be read off the data without solving, and why is it valid?

N18 — the skills

The statement

Two crews must be formed out of four technicians. Every technician has one of three skills: technicians 1 and 2 have the first, technician 3 the second, technician 4 the third. Every crew has two technicians and must have two different skills. Every technician is in one crew only. Technicians 1 and 3 do not work together. One wants to know whether a formation exists.

What is asked

Write the model and exhibit a solution. Then say what happens if technicians 2 and 3 cannot work together either, and prove it by counting.

N19 — delivering late has a price

The statement

The demands of three periods are 50, 70 and 40 units. Production costs 4 euros a unit and does not exceed 60 units per period. Holding a unit at the end of a period costs 1 euro; delivering a unit late costs 3 euros per period of delay. Everything must be delivered by the end of the third period. The minimum cost is wanted.

What is asked

Write the balance with stocks and backlog, and say why the two variables cannot both be positive at the optimum.

N20 — the hanging

The statement

Five pictures are 60, 45, 80, 50 and 70 centimetres wide. Two walls are 180 centimetres long each. Every picture must be hung, and on one wall only. One wants the free space on the two walls to be as equal as possible.

What is asked

Write the model minimising the difference between the free space of the two walls. What is the value of the LP relaxation, and why is it of no use?

S1 — the funding plan

The statement

A body must choose which projects to fund among $n \in \mathbb{Z}_{\geq 1}$ candidates. For every project $j \in \{1, 2, \dots, n\}$, the value $c_j \in \mathbb{Q}_{>0}$ is the cost, in euros, and $p_j \in \mathbb{Q}_{>0}$ is the expected return, in euros. The budget is $b \in \mathbb{Q}_{>0}$ euros. The set E contains the pairs $\{i, j\}$ of projects that use the same laboratory and cannot both be funded; the set D contains the ordered pairs (i, j) such that funding i forces funding j as well. A subset $S \subseteq \{1, 2, \dots, n\}$ gathers the strategic projects: if at least $\ell \in \mathbb{Z}_{\geq 1}$ of them are funded, the body receives an extra contribution of $\bar{r} \in \mathbb{Q}_{>0}$ euros, once only. The body wants to decide which projects to fund, at maximum total return.

S2 — the assignment with a minimum load

The statement

A department must give $n \in \mathbb{Z}_{\geq 1}$ tasks to $m \in \mathbb{Z}_{\geq 1}$ resources. For every resource $i \in \{1, 2, \dots, m\}$ and every task $j \in \{1, 2, \dots, n\}$, the value $c_{ij} \in \mathbb{Q}_{>0}$ is the cost, in euros, of giving task j to resource i , and $t_{ij} \in \mathbb{Q}_{>0}$ is the time that task takes on that resource, in hours. For every resource i , the value $b_i \in \mathbb{Q}_{>0}$ is the time available, in hours; $f_i \in \mathbb{Q}_{>0}$ is the fixed activation cost, in euros, to be paid if the resource receives at least one task; $\ell_i \in \mathbb{Q}_{>0}$ is the minimum load, in hours, that an activated resource must reach, with $\ell_i \leq b_i$. Every task goes to exactly one resource. The department wants to decide which resources to activate and how to distribute the tasks, at minimum total cost.

S3 — the coverage and the second post

The statement

An administration must cover $m \in \mathbb{Z}_{\geq 1}$ users by opening posts in some of $n \in \mathbb{Z}_{\geq 1}$ candidate sites. For every site $j \in \{1, 2, \dots, n\}$, the value $f_j \in \mathbb{Q}_{>0}$ is the opening cost, in euros, and $u_j \in \mathbb{Z}_{\geq 1}$ is the largest number of users the post can take in charge. For every site j and every user $i \in \{1, 2, \dots, m\}$, the value $d_{ij} \in \mathbb{Q}_{\geq 0}$ is the distance, in kilometres; the site may serve the user only if $d_{ij} \leq r$, where $r \in \mathbb{Q}_{>0}$ is the service radius. Every user must be taken in charge by exactly one open post. The users of a critical subset K must in addition have a second open post within the radius, even if it does not take them in charge. The administration wants to decide where to open and who takes whom in charge, at minimum total cost.

S4 — what travels together

The statement

A warehouse must store $n \in \mathbb{Z}_{\geq 1}$ objects in containers of $k \in \mathbb{Z}_{\geq 1}$ types. For every object $j \in \{1, 2, \dots, n\}$, the value $w_j \in \mathbb{Q}_{>0}$ is the weight, in kilos, and $g_j \in \{1, 2, \dots, q\}$ is the goods class, among $q \in \mathbb{Z}_{\geq 1}$ classes. For every container type $h \in \{1, 2, \dots, k\}$, the value $c_h \in \mathbb{Q}_{>0}$ is the capacity, in kilos, $e_h \in \mathbb{Q}_{>0}$ is the cost of one container of that type, in euros, and $N_h \in \mathbb{Z}_{\geq 1}$ is the number available. The set F contains the pairs $\{a, b\}$ of classes that cannot share a container. An object is not split. The warehouse wants to decide how many containers of each type to use and how to fill them, at minimum total cost.

S5 — production over n periods

The statement

A firm plans the production of one article over $n \in \mathbb{Z}_{\geq 1}$ periods. For every period $t \in \{1, 2, \dots, n\}$, the value $d_t \in \mathbb{Q}_{\geq 0}$ is the demand, in pieces; $p_t \in \mathbb{Q}_{>0}$ is the cost of making one piece, in euros; $q_t \in \mathbb{Q}_{>0}$ is the fixed set-up cost of the line, in euros, to be paid if anything is made in that period; $c_t \in \mathbb{Q}_{>0}$ is the largest quantity that can be made in the period, in pieces; $v_t \in \mathbb{Q}_{>0}$ is the smallest quantity made if anything is made, in pieces, with $v_t \leq c_t$. For every period $t \in \{1, 2, \dots, n-1\}$, the value $h_t \in \mathbb{Q}_{>0}$ is the cost of holding one piece at the end of the period, in euros; the store holds at most $M \in \mathbb{Q}_{>0}$ pieces, in every period. The values $r_0 \in \mathbb{Q}_{\geq 0}$ and $r_n \in \mathbb{Q}_{\geq 0}$ are the initial stock and the one required at the end, in pieces. The firm wants to decide how much to make in every period, at minimum total cost.

S6 — the radius to minimise

The statement

A firm must serve $m \in \mathbb{Z}_{\geq 1}$ customers by opening exactly $p \in \mathbb{Z}_{\geq 1}$ sites among $n \in \mathbb{Z}_{\geq 1}$ candidates, with $p \leq n$. For every customer $c \in \{1, 2, \dots, m\}$, the value $d_c \in \mathbb{Q}_{>0}$ is the demand, in pallets. For every site $l \in \{1, 2, \dots, n\}$, the value $u_l \in \mathbb{Q}_{>0}$ is the capacity, in pallets; for every site l and customer c , the value $t_{lc} \in \mathbb{Q}_{\geq 0}$ is the distance, in kilometres. Every customer is served entirely by one open site, and an open site does not exceed its capacity. A subset A of sites is already owned by the firm; among those not owned, at most $\bar{p} \in \mathbb{Z}_{\geq 1}$ may be opened. The firm wants to decide where to open and who serves whom, minimising the largest distance between a customer and the site serving it.

S7 — the completion of the last job

The statement

A workshop must run $n \in \mathbb{Z}_{\geq 1}$ jobs on $k \in \mathbb{Z}_{\geq 1}$ machines. For every job $j \in \{1, 2, \dots, n\}$, the value $t_j \in \mathbb{Q}_{>0}$ is the processing time, in hours, the same on every machine; $r_j \in \mathbb{Q}_{\geq 0}$ is the release date, that is the instant before which the job cannot start; $Q_j \subseteq \{1, 2, \dots, k\}$ is the set of machines qualified to run it. For every machine $i \in \{1, 2, \dots, k\}$, the value $a_i \in \mathbb{Q}_{\geq 0}$ is the instant from which the machine is free. Every machine runs one job at a time, and a job, once started, is not interrupted. The workshop wants to decide on which machine and in which order to run the jobs, minimising the instant at which the last one finishes.

S8 — the heaviest shelf

The statement

A depot must distribute $n \in \mathbb{Z}_{\geq 1}$ objects over $m \in \mathbb{Z}_{\geq 1}$ shelves. The objects belong to $q \in \mathbb{Z}_{\geq 1}$ goods classes: for every object $j \in \{1, 2, \dots, n\}$, the value $g_j \in \{1, 2, \dots, q\}$ is the class and $w_j \in \mathbb{Q}_{>0}$ is the weight, in kilos. Every shelf may hold objects of at most $s \in \mathbb{Z}_{\geq 1}$ different classes, with $s \leq q$, and carries at most $c \in \mathbb{Q}_{>0}$ kilos. The objects of a subset P are fragile and must go on shelves that are all different. The depot wants to decide the distribution, minimising the weight of the heaviest shelf.

S9 — the pieces and the waste

The statement

A workshop must obtain pieces of $k \in \mathbb{Z}_{\geq 1}$ different lengths from bars of $h \in \mathbb{Z}_{\geq 1}$ formats. For every format $f \in \{1, 2, \dots, h\}$, the value $L_f \in \mathbb{Q}_{>0}$ is the length of the bar, in metres, $g_f \in \mathbb{Q}_{>0}$ is the cost of one bar, in euros, and $N_f \in \mathbb{Z}_{\geq 1}$ is the number available. For every length $i \in \{1, 2, \dots, k\}$, the value $\ell_i \in \mathbb{Q}_{>0}$ is the length of the piece, in metres, and $b_i \in \mathbb{Z}_{\geq 1}$ is the number of pieces required. The total waste cannot exceed a fraction $e \in \mathbb{Q}_{>0}$, with $e < 1$, of the total length of the bars used. The workshop wants to decide how many bars of each format to use and how to cut them, at minimum total cost.

S10 — the flow and the arcs to activate

The statement

A logistics operator must send $Q \in \mathbb{Q}_{>0}$ units of goods from an origin node s to a destination node u over a directed network of nodes V and arcs $A \subseteq V \times V$; the nodes other than s and u are transit nodes. For every arc $(i, j) \in A$, the value $c_{ij} \in \mathbb{Q}_{>0}$ is the cost of sending one unit, in euros; $f_{ij} \in \mathbb{Q}_{>0}$ is the fixed activation cost, in euros; $k_{ij} \in \mathbb{Q}_{>0}$ is the capacity, in units. The activated arcs cannot be more than $\bar{a} \in \mathbb{Z}_{\geq 1}$, and out of every transit node at most one activated arc may leave. The operator wants to decide which arcs to activate and how much to send on them, at minimum total cost.

S11 — the staff of the service

The statement

A service runs over $n \in \mathbb{Z}_{\geq 1}$ days, on a weekly cycle. The staff is divided into $q \in \mathbb{Z}_{\geq 1}$ skills. For every day $t \in \{1, 2, \dots, n\}$ and every skill $c \in \{1, 2, \dots, q\}$, the value $b_{tc} \in \mathbb{Z}_{\geq 0}$ is the number of people of that skill who must be on duty that day. For every skill c , the value $w_c \in \mathbb{Q}_{>0}$ is the weekly cost of one person, in euros, and $m_c \in \mathbb{Z}_{\geq 0}$ is the number of people of that skill already hired. Whoever works is on duty $\ell \in \mathbb{Z}_{\geq 1}$ consecutive days and then rests $g \in \mathbb{Z}_{\geq 1}$, and the cycle may start on any day. The service wants to decide how many people to hire per skill and with which starting day, at minimum total cost.

S12 — the funds to buy

The statement

An investor has a capital of $K \in \mathbb{Q}_{>0}$ euros and chooses among $n \in \mathbb{Z}_{\geq 1}$ funds, divided into $k \in \mathbb{Z}_{\geq 1}$ sectors. For every fund $j \in \{1, 2, \dots, n\}$, the value $s_j \in \{1, 2, \dots, k\}$ is the sector, $q_j \in \mathbb{Q}_{>0}$ is the lot size, in euros — the fund is bought in whole lots only — and $r_j \in \mathbb{Q}_{>0}$ is the expected return per euro invested. In no sector may more than a fraction $a \in \mathbb{Q}_{>0}$, with $a \leq 1$, of the capital be invested. At least $p \in \mathbb{Z}_{\geq 1}$ different funds must be chosen, and a chosen fund receives at least $v_j \in \mathbb{Z}_{\geq 1}$ lots. The investor wants to decide how many lots of each fund to buy, at maximum total return.

S13 — the canteen and the stocks

The statement

A canteen prepares the ration of $n \in \mathbb{Z}_{\geq 1}$ days using $s \in \mathbb{Z}_{\geq 1}$ foods and respecting $r \in \mathbb{Z}_{\geq 1}$ nutritional constraints. For every food $i \in \{1, 2, \dots, s\}$, the value $w_i \in \mathbb{Q}_{>0}$ is the cost of one kilo, in euros; $c_i \in \mathbb{Q}_{>0}$ is the smallest quantity, in kilos, used on a day where the food appears; $u_i \in \mathbb{Q}_{>0}$ is the largest daily quantity, with $c_i \leq u_i$; and $h_i \in \mathbb{Q}_{>0}$ is the cost of holding one kilo in the pantry from one day to the next, in euros. For every food i and nutrient $j \in \{1, 2, \dots, r\}$, the value $g_{ij} \in \mathbb{Q}_{\geq 0}$ is the amount of nutrient j in one kilo of food i ; for every nutrient j , every day at least $a_j \in \mathbb{Q}_{>0}$ and at most $b_j \in \mathbb{Q}_{>0}$ are needed. Every day at least $t \in \mathbb{Z}_{\geq 1}$ different foods must appear. The canteen wants to decide how much to buy and how much to use each day, at minimum total cost.

S14 — the order of the jobs

The statement

On a single machine $n \in \mathbb{Z}_{\geq 1}$ jobs must run. For every job $j \in \{1, 2, \dots, n\}$, the value $t_j \in \mathbb{Q}_{>0}$ is the processing time, in hours, and $d_j \in \mathbb{Q}_{>0}$ is the due date, that is the instant by which the job should be completed; delivering late costs $w_j \in \mathbb{Q}_{>0}$ euros per hour of delay. For every pair of distinct jobs i and j , the value $s_{ij} \in \mathbb{Q}_{\geq 0}$ is the set-up time, in hours, needed to go from job i to job j . The machine starts at time 0, runs one job at a time and does not stop during a job. The workshop wants to decide the order of the jobs, minimising the total cost of the delays.

S15 — covering the shifts

The statement

A service company must cover $m \in \mathbb{Z}_{\geq 1}$ zones for $n \in \mathbb{Z}_{\geq 1}$ periods, with $k \in \mathbb{Z}_{\geq 1}$ crews. For every crew $j \in \{1, 2, \dots, k\}$, the value $c_j \in \mathbb{Q}_{>0}$ is the cost of employing it for one period, in euros, and $S_j \subseteq \{1, 2, \dots, m\}$ is the set of zones it can cover. In every period every zone must be covered by at least one employed crew, and a crew employed in a period covers all and only the zones of S_j . A crew cannot be employed for more than $\ell \in \mathbb{Z}_{\geq 1}$ periods out of n , and if it is employed it is employed for at least $v \in \mathbb{Z}_{\geq 1}$ periods, with $v \leq \ell$. The company wants to decide which crews to employ and in which periods, at minimum total cost.

S16 — assigning the theses

The statement

A degree programme must assign $n \in \mathbb{Z}_{\geq 1}$ students to $m \in \mathbb{Z}_{\geq 1}$ supervisors. For every student $i \in \{1, 2, \dots, n\}$ and every supervisor $j \in \{1, 2, \dots, m\}$, the value $v_{ij} \in \mathbb{Z}_{\geq 0}$ is the preference score the student gives that supervisor. For every supervisor j , the value $u_j \in \mathbb{Z}_{\geq 1}$ is the largest number of students they can follow. Every student is assigned to exactly one supervisor, and nobody may receive a supervisor to whom they gave a score below $\bar{v} \in \mathbb{Z}_{\geq 0}$. The programme wants to decide the assignment maximising the sum of the preferences, with the constraint that the difference between the number of students of the busiest supervisor and that of the least busy among those with at least one student does not exceed $g \in \mathbb{Z}_{\geq 0}$.

S17 — when to stop the machines

The statement

A plant has $n \in \mathbb{Z}_{\geq 1}$ machines and plans the maintenance over $T \in \mathbb{Z}_{\geq 1}$ periods. For every machine $j \in \{1, 2, \dots, n\}$, the values $a_j \in \{1, 2, \dots, T\}$ and $b_j \in \{1, 2, \dots, T\}$, with $a_j \leq b_j$, delimit the window in which the maintenance must be done, once only; $p_j \in \mathbb{Q}_{>0}$ is the hourly output of the machine, in pieces; $\ell_j \in \mathbb{Z}_{\geq 1}$ is the duration of the maintenance, in consecutive periods. For every machine j and period t , the value $c_{jt} \in \mathbb{Q}_{>0}$ is the cost of the maintenance crew, in euros, if the maintenance starts in that period. In no period may more than $k \in \mathbb{Z}_{\geq 1}$ machines be stopped, and the total lost output cannot exceed $M \in \mathbb{Q}_{>0}$ pieces. The plant wants to decide when to stop each machine, at minimum total cost.

S18 — two groups, few links

The statement

An organisation must split into two teams the $n \in \mathbb{Z}_{\geq 1}$ nodes of an undirected graph with edge set E . For every node $i \in \{1, 2, \dots, n\}$, the value $w_i \in \mathbb{Q}_{>0}$ is the workload it carries; for every edge $\{i, j\} \in E$, the value $u_{ij} \in \mathbb{Q}_{>0}$ is the intensity of the exchange between the two nodes. The total load of each team must lie between $L \in \mathbb{Q}_{>0}$ and $U \in \mathbb{Q}_{>0}$, with $L \leq U$. Some pairs of nodes, gathered in the set P , must end up in the same team; others, gathered in F , in different teams. The organisation wants to decide the split, minimising the total intensity of the exchanges between the two teams.

S19 — what goes on what

The statement

A carrier must ship $n \in \mathbb{Z}_{\geq 1}$ loads with $m \in \mathbb{Z}_{\geq 1}$ vehicles. For every load $j \in \{1, 2, \dots, n\}$, the value $w_j \in \mathbb{Q}_{>0}$ is the weight, in kilos, $o_j \in \mathbb{Q}_{>0}$ is the volume, in cubic metres, and $V_j \subseteq \{1, 2, \dots, m\}$ is the set of vehicles it may go on. For every vehicle $i \in \{1, 2, \dots, m\}$, the value $c_i \in \mathbb{Q}_{>0}$ is the payload, in kilos, $s_i \in \mathbb{Q}_{>0}$ is the volume available, in cubic metres, and $f_i \in \mathbb{Q}_{>0}$ is the cost of using it, in euros. The set E contains the pairs $\{a, b\}$ of loads that cannot travel on the same vehicle. Every load goes on exactly one vehicle. The carrier wants to decide which vehicles to use and how to split the loads, at minimum total cost.

S20 — the technologies and the brackets

The statement

An operator must install a total capacity of at least $Q \in \mathbb{Q}_{>0}$ megawatts choosing among $n \in \mathbb{Z}_{\geq 1}$ technologies. For every technology $j \in \{1, 2, \dots, n\}$, the value $k_j \in \mathbb{Q}_{>0}$ is the capacity of one installed unit, in megawatts, and the units are installed in whole numbers; the unit cost falls in brackets: for every bracket $h \in \{1, 2, \dots, m_j\}$, with $m_j \in \mathbb{Z}_{\geq 1}$, the value $u_{jh} \in \mathbb{Z}_{\geq 1}$ is the number of units in the bracket and $c_{jh} \in \mathbb{Q}_{>0}$ is the cost of one unit in that bracket, with $c_{j1} > c_{j2} > \dots > c_{jm_j}$; a bracket is used only once the previous ones are full. At most $p \in \mathbb{Z}_{\geq 1}$ different technologies may be used, and a technology that is used is installed for at least $v_j \in \mathbb{Z}_{\geq 1}$ units. The operator wants to decide how much of each technology to install, at minimum total cost.

Organisation of the course

The format of every exercise — and of the exam

1. **Model.** Variables (with their count), objective, constraints with their description; and if two families are linked, the **proof of the link in both directions**.
2. **Instance.** The model written out in full on a small numerical instance.
3. **Heuristic.** A constructive algorithm, run step by step, giving the bound on the primal side.
4. **Dual.** The dual of the LP relaxation — general and on the instance — and a feasible dual solution built by hand.

The thread through the course

$LB \leq z(MILP) \leq UB$: a model is not merely written, it is **tightened from both sides** before being handed to the solver.

Marking criteria

Dimension	Weight
Correctness of the model (variables, objective, constraints)	35%
Proof of the link between the variables (both directions)	25%
Constructive heuristic and its correct execution	20%
Dual of the relaxation and a feasible dual solution	20%

Typical discussion questions

Is the link constraint aggregated or disaggregated, and what difference does it make to the relaxation? Is the opposite direction of the implication imposed by the constraint or does it follow from optimality? What is the smallest big-M the data justify? Does the heuristic find the optimum, and how does one know without the solver?

The eight most common mistakes

1. Writing an implication whose conclusion is true anyway (vacuous contrapositive).
2. Proving only one direction of an implication "imposed by the constraint": both are always needed.
3. Confusing "imposed by the constraint" with "follows from optimality": the second needs the exchange argument, "one can see it pays off" is not enough.
4. Concluding "in every optimum" when the coefficient is only ≥ 0 : the correct conclusion is "there exists an optimum in which...".
5. Using a huge big-M "to be safe": it weakens the relaxation for nothing.
6. Forgetting that $A + B \leq 1$ does **not** imply that at least one of the two is 1.
7. Getting the direction or the sign of a dual variable wrong with respect to the primal constraint.
8. Building a dual solution and not checking its feasibility on every constraint.

The material of the course

- **Site** fabiofurini.github.io/mip-modelling — the notes online, with the code embedded in every page.
- **Notebooks in Colab**: one per model, opened from the badge at the top of each page and running in the browser. The licence shipped with gurobipy is enough for every instance of the course.
- **Code** at github.com/fabiofurini/mip-modelling, under the MIT licence; text, figures and data under CC BY 4.0.
- Italian version: fabiofurini.github.io/modellazione-mip.

Enjoy the course.